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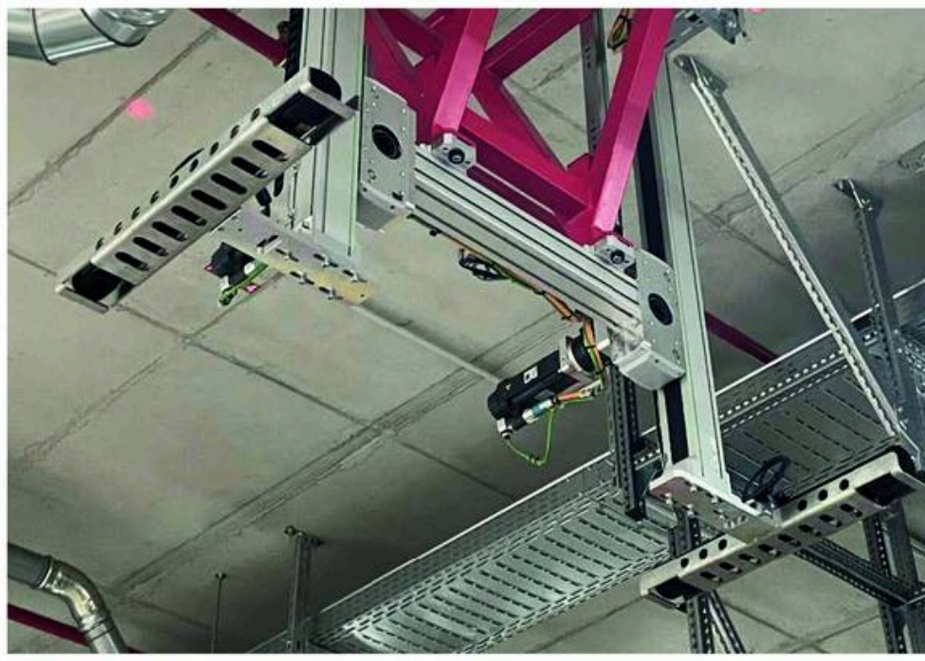

Advanced Order Picking System

7 types of load carriers and
cardboard boxes, 250 picks/hour

USE CASE

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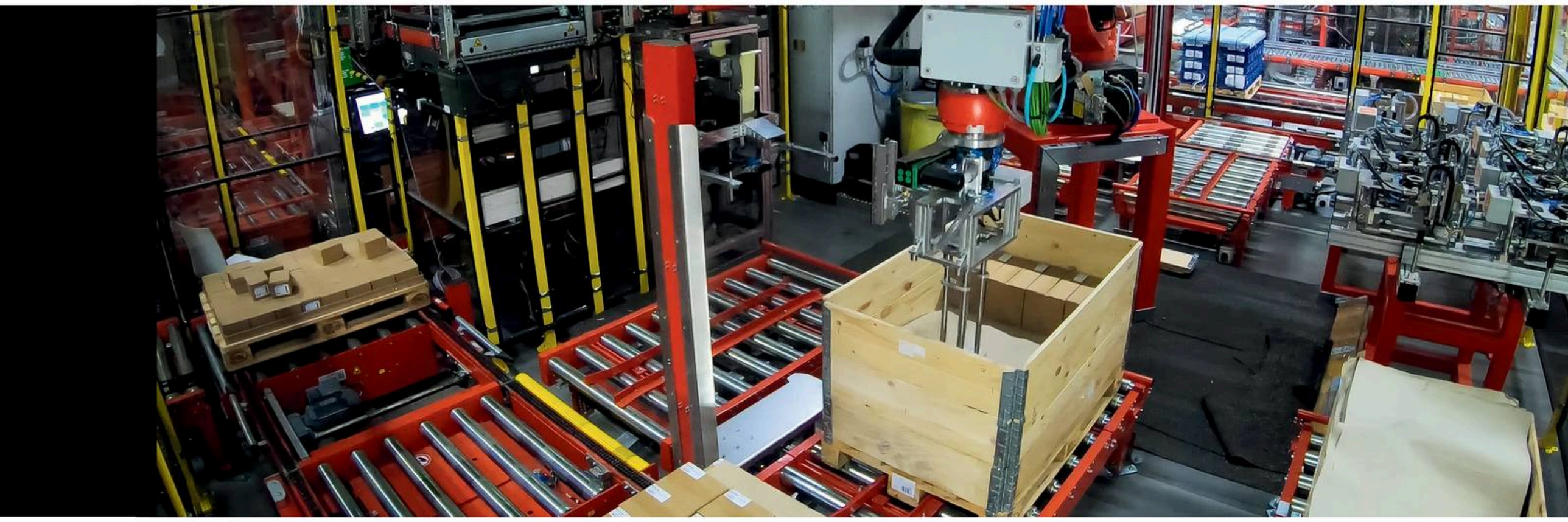
Overview

How do you teach a robot to see not just a stack of boxes, but the thin slip sheet between the layers and the stretch band holding them together?

EEP Robotics' client needed to automate the depalletizing of a wide variety of cardboard boxes and plastic load carriers (KLTs) from Euro pallets. In the end, this solution won them the 1st prize at our Photoneo Technology & Applications Conference 2024.

By building their solution around four of our PhoXi 3D Scanners L, EEP engineered a truly intelligent and autonomous order picking system.

Partner: EEP-Robotics GmbH
Industry: Logistics & Order Fulfillment
Application: Order Picking System
Technology: PhoXi 3D Scanners with a Hybrid Software



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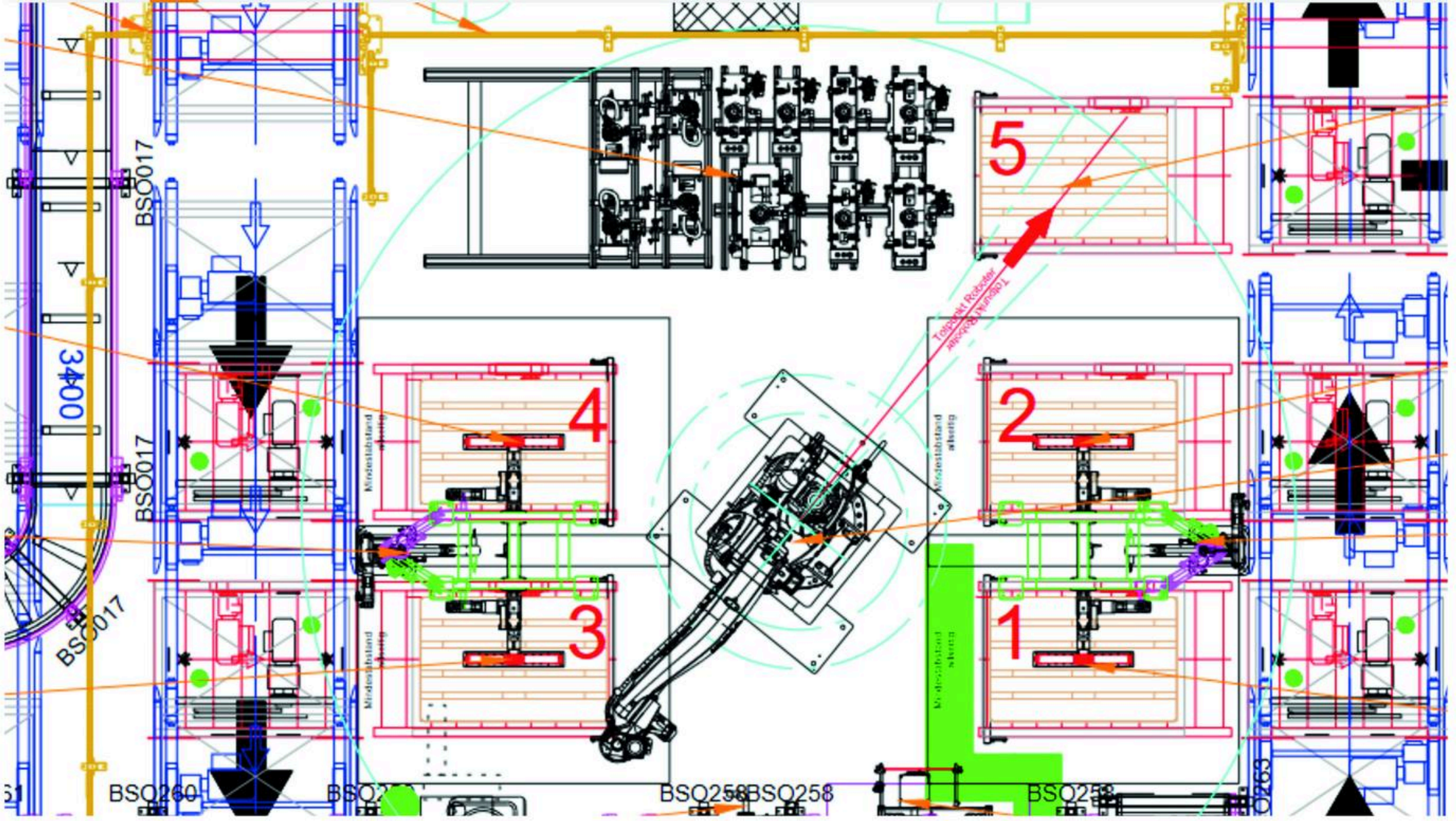

The Challenge

Automating Complex, Unstructured Depalletizing

- Unstructured Pallets: Vision-based identification of randomly arranged items and their joints for precise localization.
- Object Variety: Capability to handle seven distinct box and carrier types across various dimensions.
- Complex Material Handling: Detection and management of intermediate layers, wooden frames, and stretch bands.
- Extended Scan Volume: High-resolution, full-height scanning for pallets exceeding standard sensor ranges.
- High Throughput: Minimum performance of 250 picks per hour, including ten robotic gripper changes.

PhoXi 3D Scanners with a Hybrid Software Approach

Four PhoXi 3D Scanner L units were mounted on a ceiling servo axis system. This innovative mechanical setup allows the scanners to move vertically, effectively extending their scanning range to cover the entire pallet height while maintaining the high resolution needed for precise position detection.

The core of the solution lies in a unique, hybrid software architecture that combines the strengths of Photoneo's native software with EEP's custom AI algorithms running directly on our scanner's controller.

The Solution

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PhoXi 3D Scanners with a Hybrid Software Approach

- **Photoneo's Onboard Software:** Our powerful, built-in object recognition software was used for its "plug & play" capability to reliably detect and locate all cardboard boxes and plastic carriers on the top layer.
- **EEP's Custom Docker Application:** A key decision factor for EEP was the ability to run their own custom software in a Docker container on the Photoneo controller. This EEP-developed software processes the raw point cloud to perform specialized tasks, including the detection of intermediate cardboard layers, wooden pallet frames, and stretch bands.

The Solution

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Architecture

1. **Trigger from PLC:** The process begins when the main robotic cell controller (EEP PLC) sends a "trigger" command to start the vision task.
2. **EEP's Software Receives Command:** The command goes to EEP's custom software environment, which acts as the central brain. This software parses the trigger to understand what the PLC is asking for.
3. **Standard Photoneo Scan:** EEP's software then triggers the standard Photoneo application. The scanner performs a scan and uses its built-in pipeline to quickly detect standard objects like boxes. These results are immediately sent back to EEP's software.
4. **Simultaneous Raw Data Access:** In parallel, a custom function allows EEP's software to access the raw 3D point cloud data generated by the scan.
5. **Custom AI Processing:** EEP's software takes this raw data and runs its own advanced algorithms (using technologies like TensorFlow and PyTorch) to perform specialized detections for items like intermediate layers and stretch bands.
6. **Combine and Process Results:** Finally, EEP's software combines the standard results from Photoneo's pipeline with the custom results from its own analysis. It processes this unified data into a single, comprehensive command set and sends it back to the EEP PLC.

A Step by Step Order Picking Workflow

The floor operates a single robot that intelligently alternates between two parallel floor "spheres": one for handling boxes and the other for handling KLTs (Small Load Carriers).

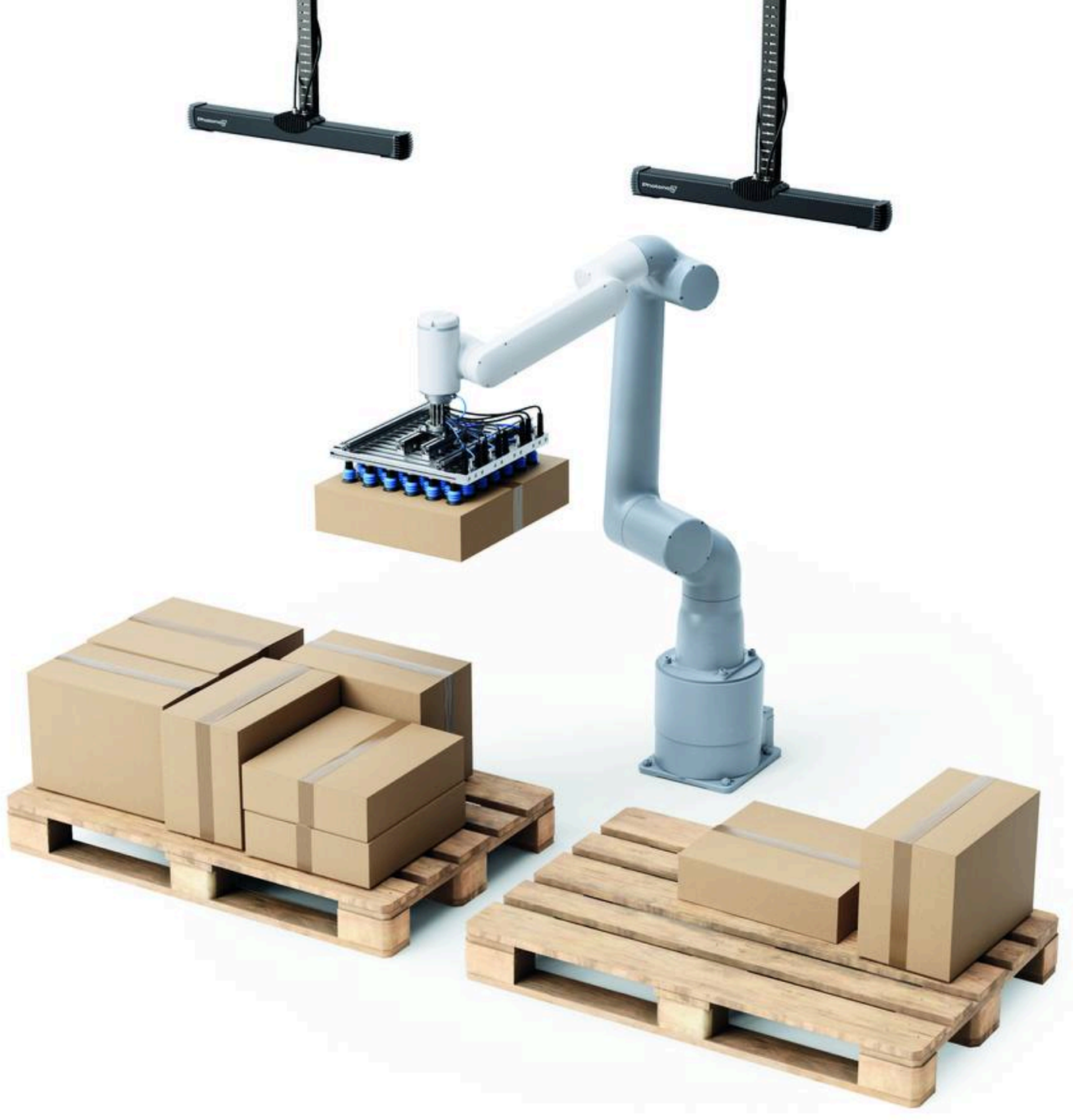
The robot works in perfect synchronization with Photoneo 3D scanners and autonomously changes grippers to cover all the jobs entailed in this complex application.

The Workflow

Step by Step

1. **Automatic delivery of the pallet in the robot cell:** The process begins with the pallet being automatically delivered and positioned inside the robot's working cell, ready for processing.
2. **Pallet scanning with PhoXi 3D scanner on servo axis:** A PhoXi 3D scanner mounted on a servo axis performs a detailed scan of the pallet and its contents. This scan provides the robot with the exact location, orientation, and dimensions of the items to be picked.
3. **Grab the right gripper with the Kuka robot:** Based on the scanning results and the task (handling a box or a KLT/tray), the Kuka robot autonomously selects and equips the appropriate gripper from its tool changer.
4. **Pick the box or tray:** The robot uses the equipped gripper to pick the specified box or tray from the pallet.
5. **Labelling of the box or tray:** The picked item undergoes a labelling process to identify its contents and destination.
6. **Reading the RFID Tag of the tray:** If the item is a tray, its RFID tag is read to confirm its identity and track its progress through the system.
7. **Place the product in the tote provided on the customer conveyor system:** The final product is carefully placed into a tote or container that is located on the customer's conveyor system for further transport or processing.
8. **Automatic delivery of the pallet in the robot cell:** After the product is placed in the tote, the cycle completes, bringing us back to the start - ready for the next pallet delivery or continuing with the current pallet until all items are processed.

A Prize-Winning Advantage



250 picks per hour and accurately handles the diverse range of products and pallet.

Ease of Use & Powerful Customization

Well-trained vision system that can be deployed in other applications

Consistent and reliable solution that works without surprises

Results And Benefits

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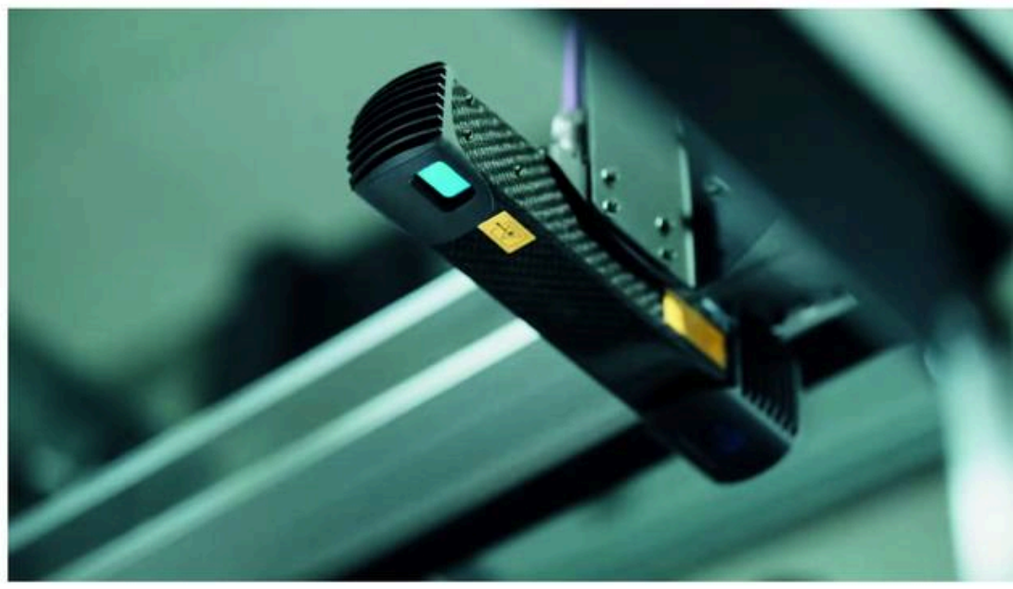
The World's Fastest Pocket Sorter Induction



800-1000 Picks Per Hour

Up To 99.7% Accuracy

USE CASE

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Overview

In summer 2024, Photoneo's advanced 3D vision technology became the cornerstone of an intralogistics automation project that created the world's fastest pocket sorter induction solution.

Through strategic partnership with Rebl Industries, our MotionCam-3D technology enabled unprecedented throughput rates of 800-1000 picks per hour with >99.7% accuracy at one of the Nordic region's busiest retail distribution centers.

Client: Retail and Logistics Facility (via Rebl Industries AB)
Industry: Retail, Logistics & Warehousing
Application: Intralogistics, Pocket Sorter Induction
Technology: MotionCam-3D

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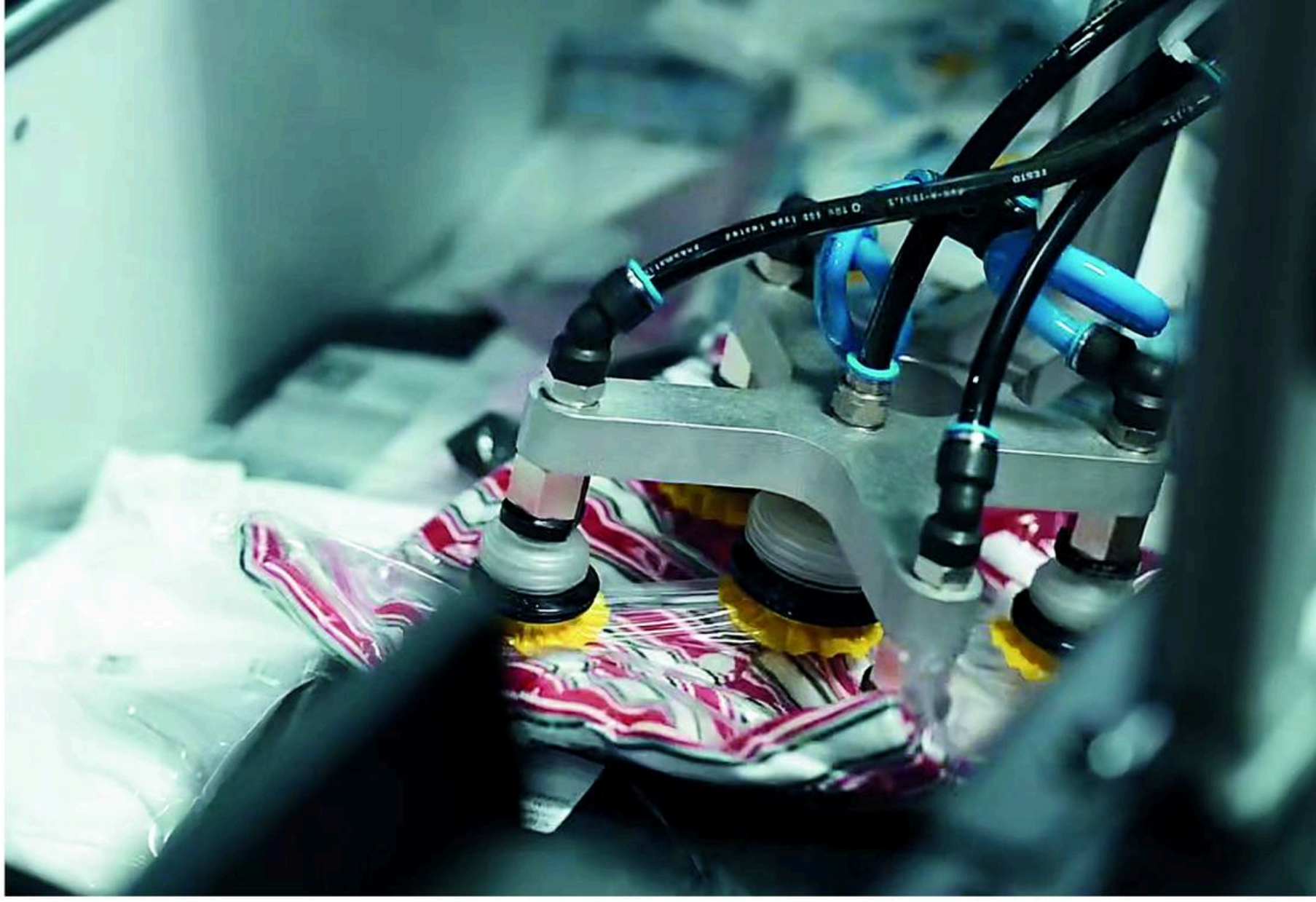
The Challenge

Vision-Powered Automation at Scale

The Nordic retail giant's existing pocket sorter infrastructure faced the classic automation challenge: how to reliably handle diverse, unpredictable objects at industrial speeds.

Multiple parallel sorter lines, each previously requiring dedicated human operators, needed to process thousands of mixed SKUs daily - varying in size, shape, material, and packaging.



Visual intelligence required to:

- Accurately locate objects in cluttered, dynamic environments
- Distinguish between different product types without pre-programming
- Adapt to new SKUs without extensive system reconfiguration
- Maintain precision at high operational speeds

The Requirements

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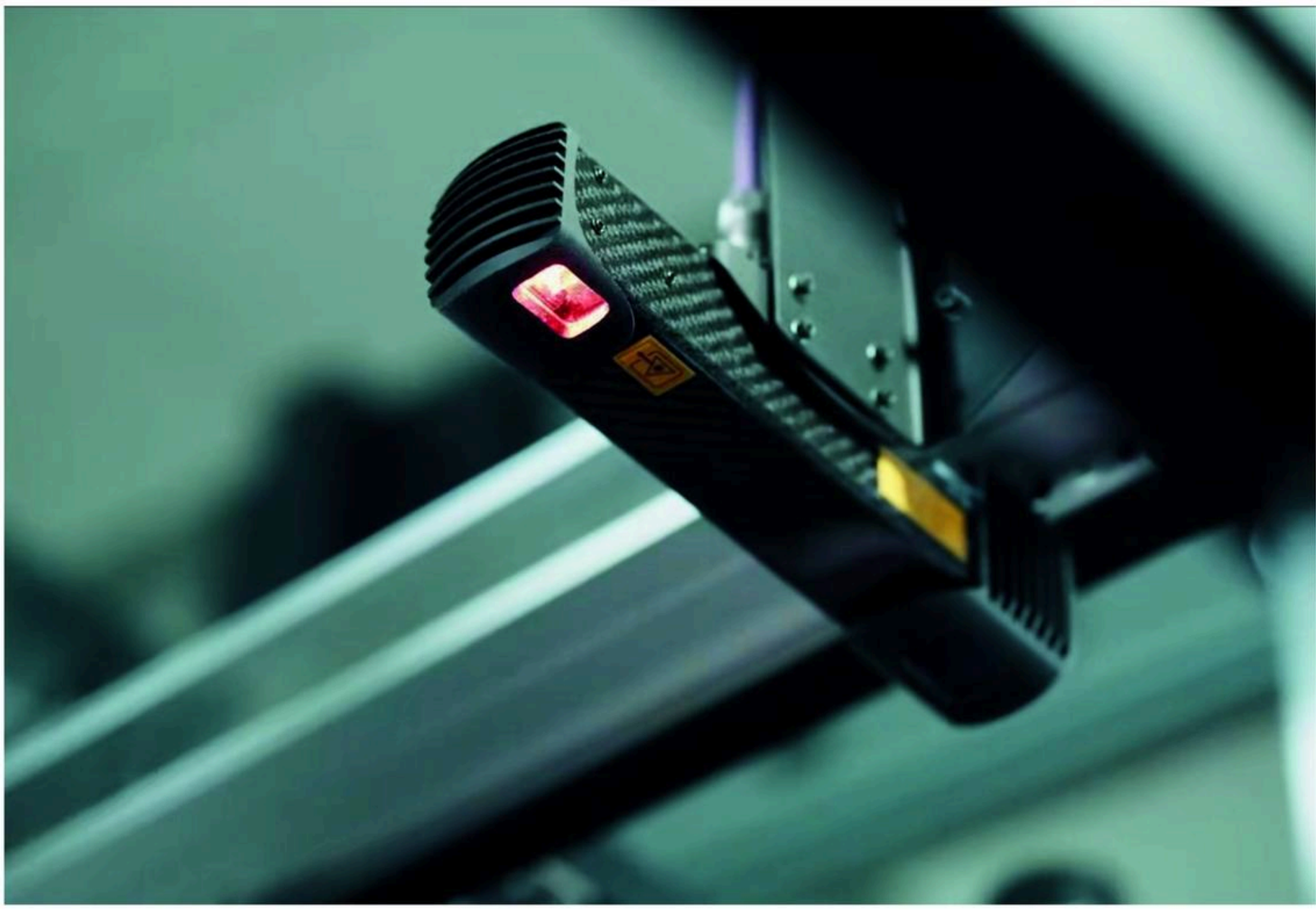
3D Vision Intelligence



This is where Photoneo's 3D vision technology became essential—providing the "eyes" that would enable Rebl Industries' robotic systems to achieve human-like adaptability at superhuman speeds.

Our contribution to this record-breaking system centered on delivering comprehensive visual intelligence that could match the complexity demands of modern fulfilment operations.

The Photoneo Solution

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How it works

MotionCam-3D Color M

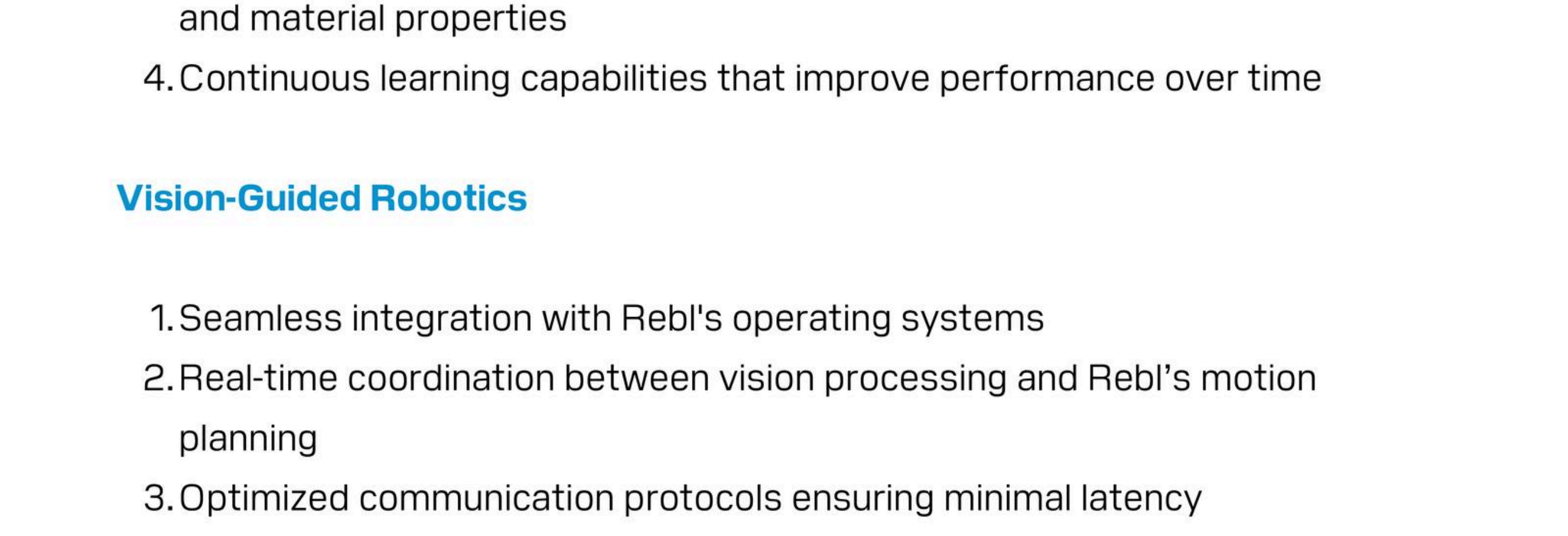
1. High-speed 3D scanning capability essential for tracking objects in fast-moving environments
2. Advanced depth perception enabling precise localization in cluttered bins
3. Integrated color information for enhanced object classification
4. Sub-millimeter accuracy ensuring reliable grasp point identification

AI AnyPick Technology

1. Deep learning algorithms trained to recognize and handle diverse products
2. Real-time adaptation to new SKUs without manual programming
3. Intelligent grasp point selection optimized for each unique item geometry and material properties
4. Continuous learning capabilities that improve performance over time

Vision-Guided Robotics

1. Seamless integration with Rebl's operating systems
2. Real-time coordination between vision processing and Rebl’s motion planning
3. Optimized communication protocols ensuring minimal latency

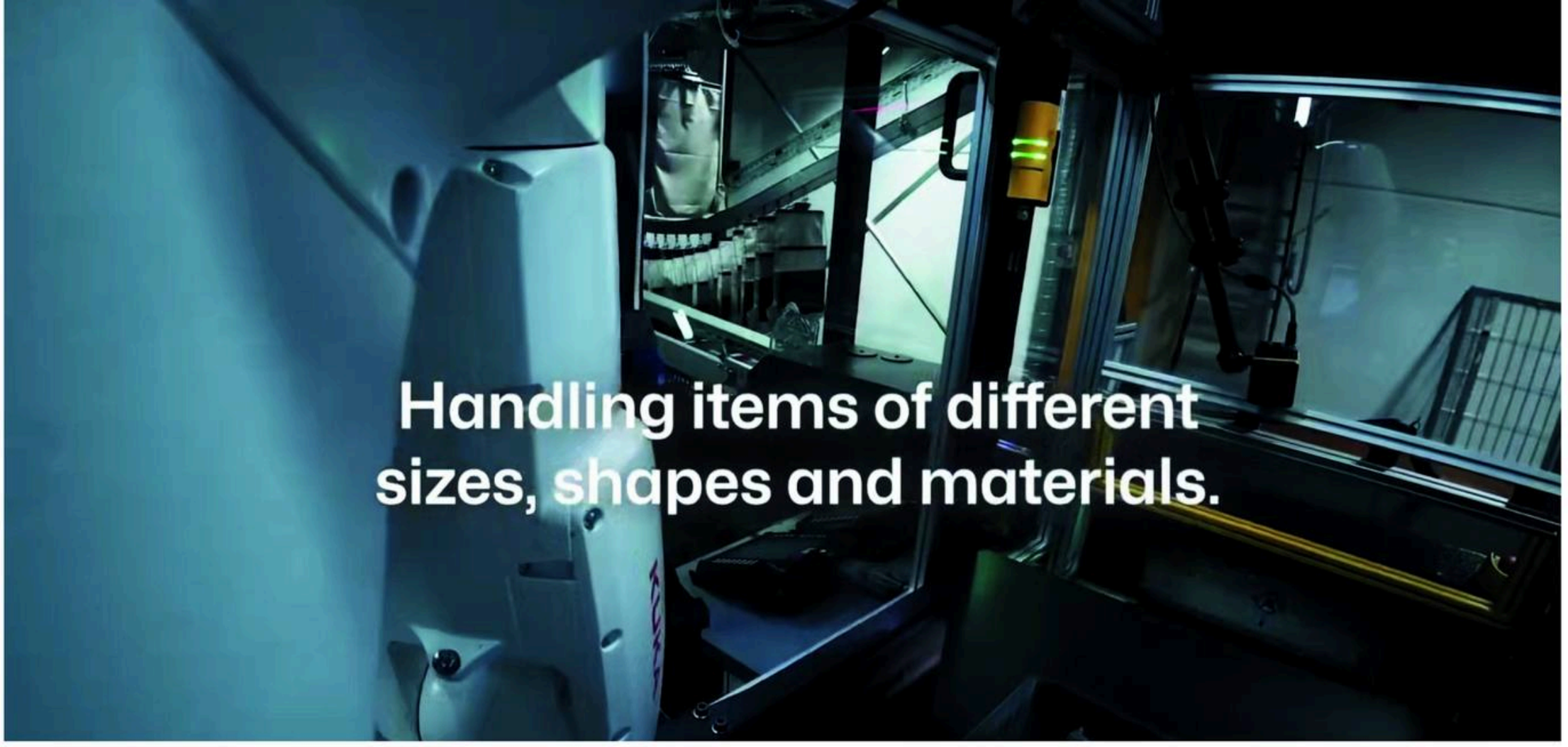


The Implementation

Photoneo

Vision Intelligence Layer

- MotionCam-3D cameras providing real-time 3D point cloud data
- AI AnyPick processing visual information for object recognition
- Continuous AI re-training and optimization based on operational data

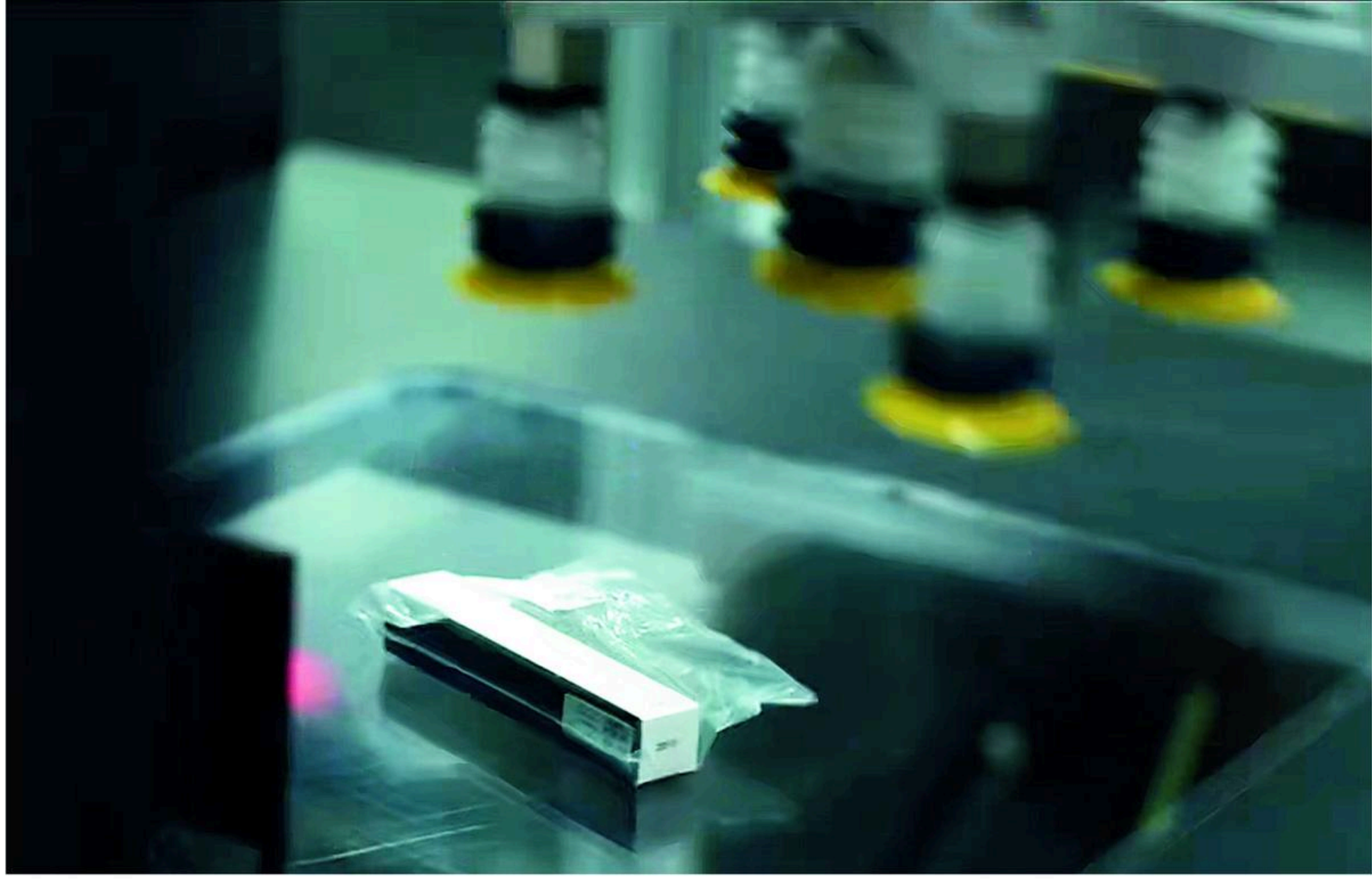


Rebl Industries

Automation Platform

- Robot-as-a-Service (RAAS) model enabling immediate deployment
- Proprietary motion planning software optimizing robot trajectories
- Comprehensive logistics software including AI platform integration
- Custom workstation design requiring no facility modifications
- 2 KUKA industrial robots with specialized vacuum grippers
- Complete tracking infrastructure with 6 Zebra Technologies scanners

The industry-leading results



99.7%+ Uptime & Stability,
800-1000 picks per hour

Reduced labour from
4 workers to 1

Handles 100% of
existing SKUs

30%+ increase in overall throughput capacity
& 40% reduction in operating costs

Results And Benefits

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3D Vision Powers AI Robotics In Smart Palletizing

450 Mixed Pieces / Hour

Over 99% uptime and stability

USE CASE

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Overview

A global retail customer in Sweden faced a significant ergonomic challenge with the manual handling of nearly two million uniquely sized parcels annually. To automate this heavy-duty task, automation partner Rebl Industries developed Rebl Palletizer, an AI-powered robotic workmate.

The critical challenge was enabling the robot to handle a constant flow of random, unknown box sizes without stopping the line, where Photoneo played a key role with the industry-leading MotionCam-3D that works smoothly with Rebl's Vision Guided Robotics (VGR) software, together serving as the "eyes" and "spatial intelligence" of the system.

Client: Retail and Logistics Facility (via Rebl Industries AB)
Industry: Retail, Logistics & Warehousing
Application: Vision Guided Robotics, Palletizing
Technology: MotionCam-3D

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The Challenge

Automating Mixed-Case Palletizing in Motion

The "wrap" station at the Global Retail Customer is one of the most physically demanding jobs in the facility. Co-workers manually lift and stack parcels, weighing up to 20 kg each, from a sorting line into delivery cages. During a single shift, one person might handle over 1,000 packages.

The primary obstacle to automation was variability. To minimize waste and shipping costs, the Global Retail Customer creates made-to-order boxes tailored to each product. This means the robotic cell would face a continuous stream of parcels with different dimensions, shapes, and weights.

The solution required a complex system

Identify the precise 3D dimensions and orientation of each unique box.

Operate at high speed (up to 450 pieces per hour) without stopping.

Provide accurate data to an AI for intelligent stacking and path planning.

Ensure reliable and safe gripping for objects up to 35 kg.

The Requirements

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High-Fidelity Vision for AI-Powered Robotics

While Rebl Industries provided the AI-driven robot in a flexible Robot-as-a-Service model, and Zebra Technologies supplied barcode scanners for sorting, the entire system hinged on one crucial capability: perception.

The robot had to see, understand, and interact with the random flow of goods in real-time. This is where Photoneo's technology became the core of the solution.

The Photoneo Solution

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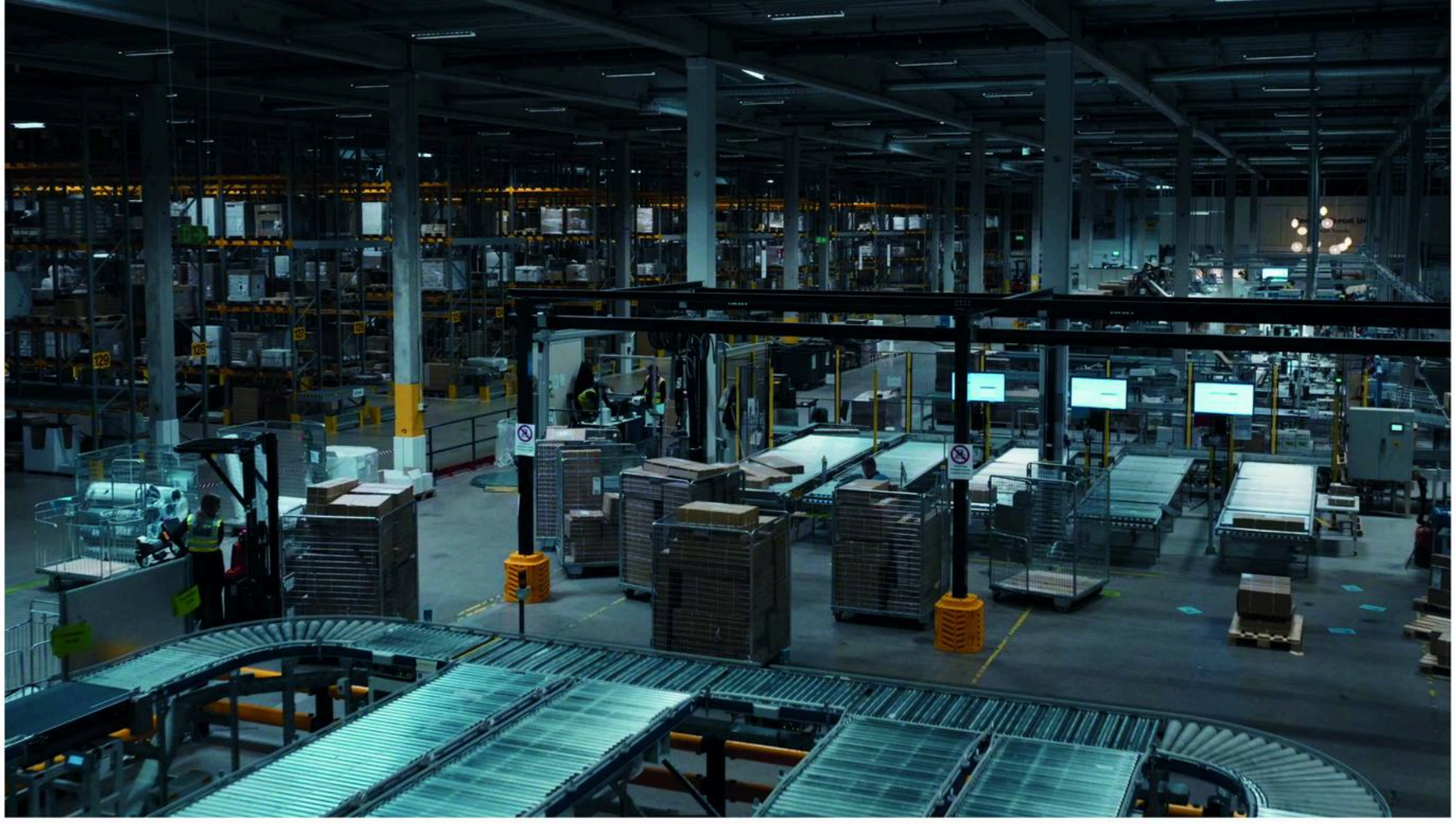
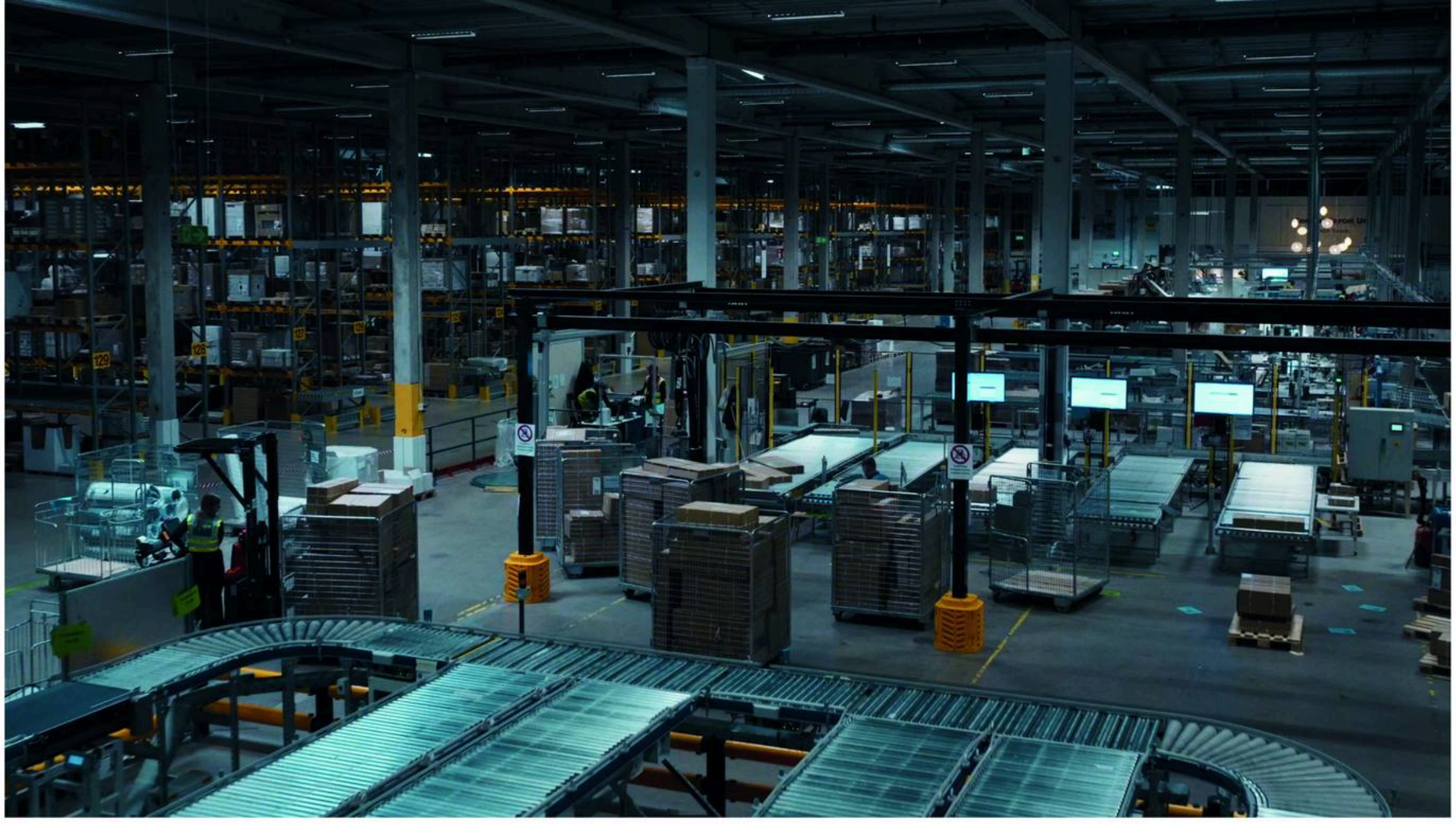


How It Works

Capturing Reality in Motion with MotionCam-3D

We provide our patented MotionCam-3D over the conveyor line. This scanner is unique in its ability to capture high-resolution, high-accuracy 3D point clouds of objects moving at high speeds.

As parcels travel down the line, the MotionCam-3D scans the scene for 3D data of each one, capturing its precise dimensions, volume, and position—all without ever needing the conveyor to stop.



The System Workflow

The complete workflow, powered by Photoneo's 3D vision

- A parcel of any size enters the robotic cell on the conveyor.
- A Zebra FS40 barcode scanner reads its label, providing routing information.
- As it passes under the Photoneo MotionCam-3D, a detailed 3D scan is generated.
- This crucial 3D data is passed to the Rebl Workmate's AI.
- Rebl's generative stacking AI uses the precise dimensions from our scan to determine the ideal placement on the pallet for maximum stability and density.
- Rebl's robot, guided by our vision software, executes a perfect pick-and-place operation.

The solution required a complex system

I99%+ Uptime & Stability

450 Pcs/Hour Throughput

Real-Time 3D Vision In Motion

Zero-Stop Conveyor Automation

The Results And Benefits

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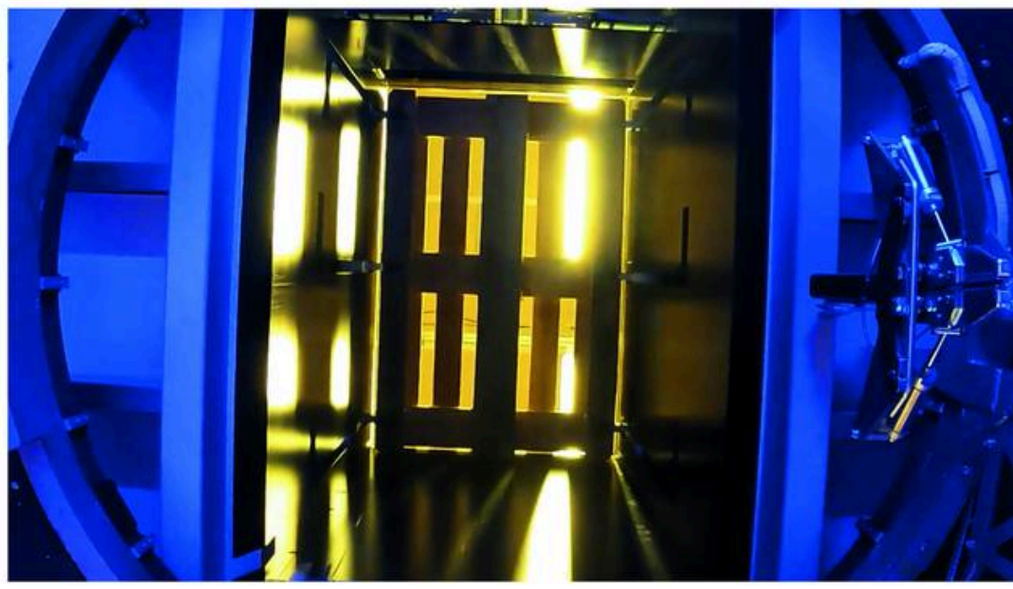
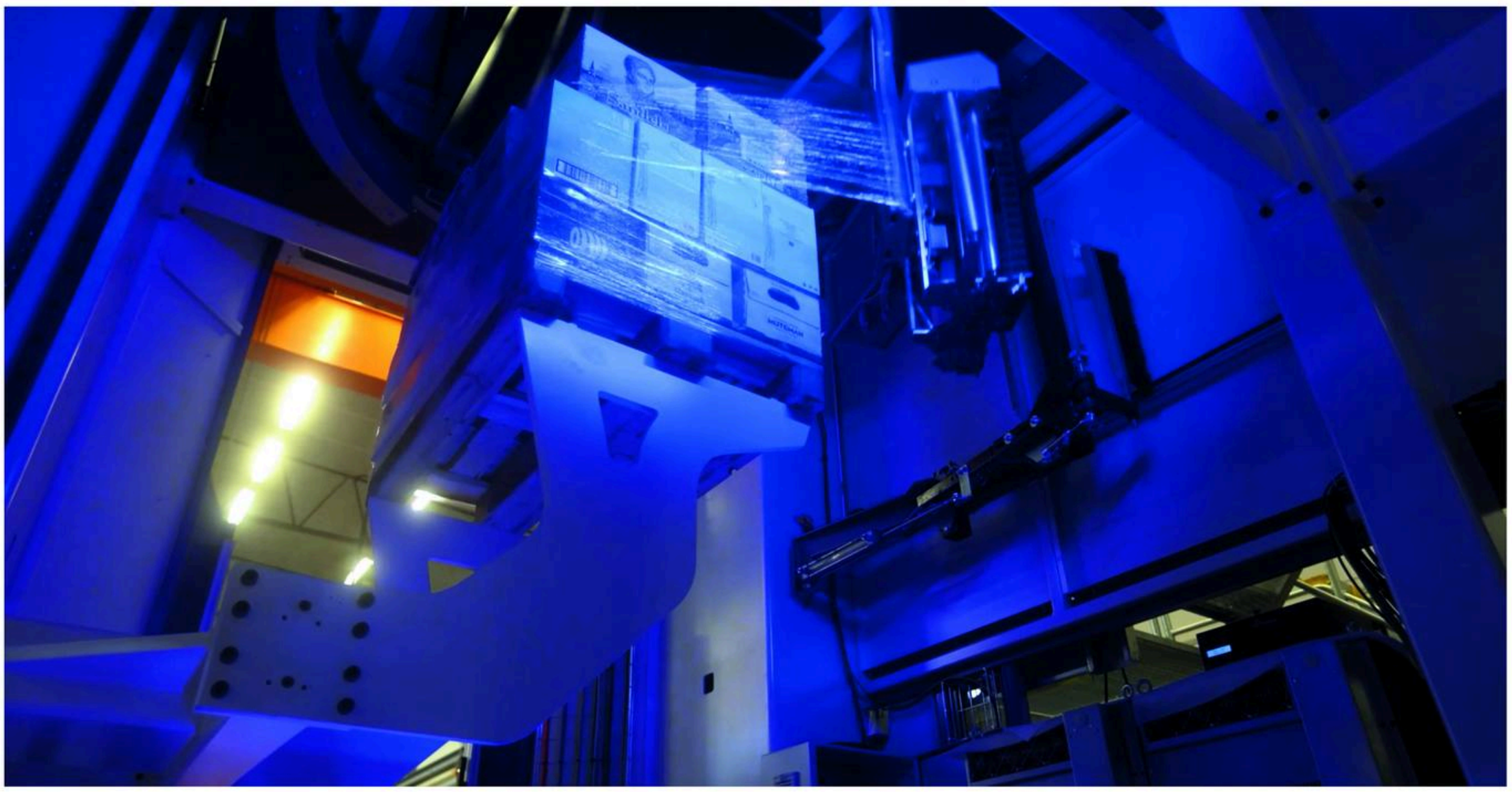
850 Boxes/Hour. Unmatched Accuracy. In Motion.



Mixed Palletizing with Scalable Automation



USE CASE

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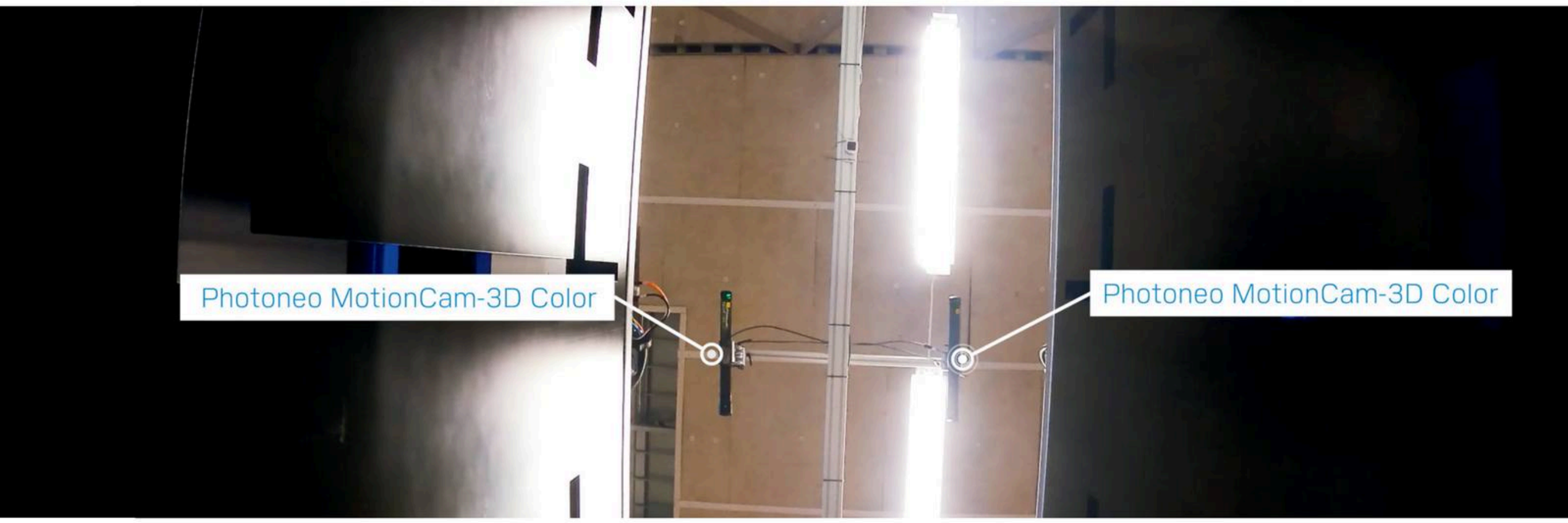
The Challange

Imagine countless boxes of various shapes, sizes, weights, and materials, impeccably racing through a conveyor belt. Nonstop. No human interaction needed, no extra space filled - that's usually not the case. Traditional palletizing, especially for mixed products, often involves extensive manual labor and complex logistics.

Labor shortages, worker safety concerns from repetitive lifting, inefficient pallet space use, and difficulty scaling hinder operations. Manual processes also lack the precision needed for stable pallets, risking product damage and impacting customer satisfaction.

Partner: Yaskawa
Industry: Logistics & Warehousing
Application: Palletizing & Depalletizing
Technology: MotionCam-3D



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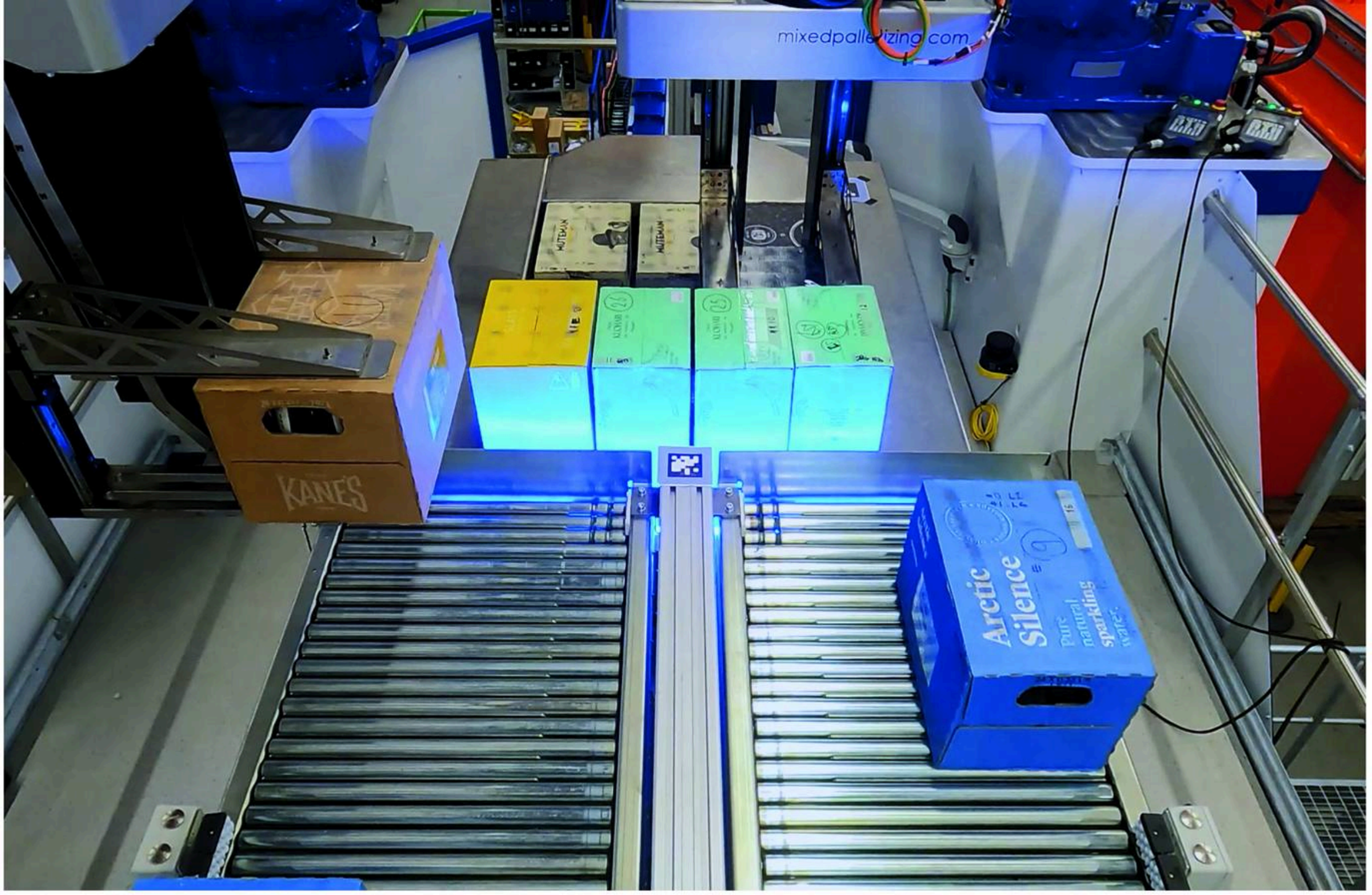
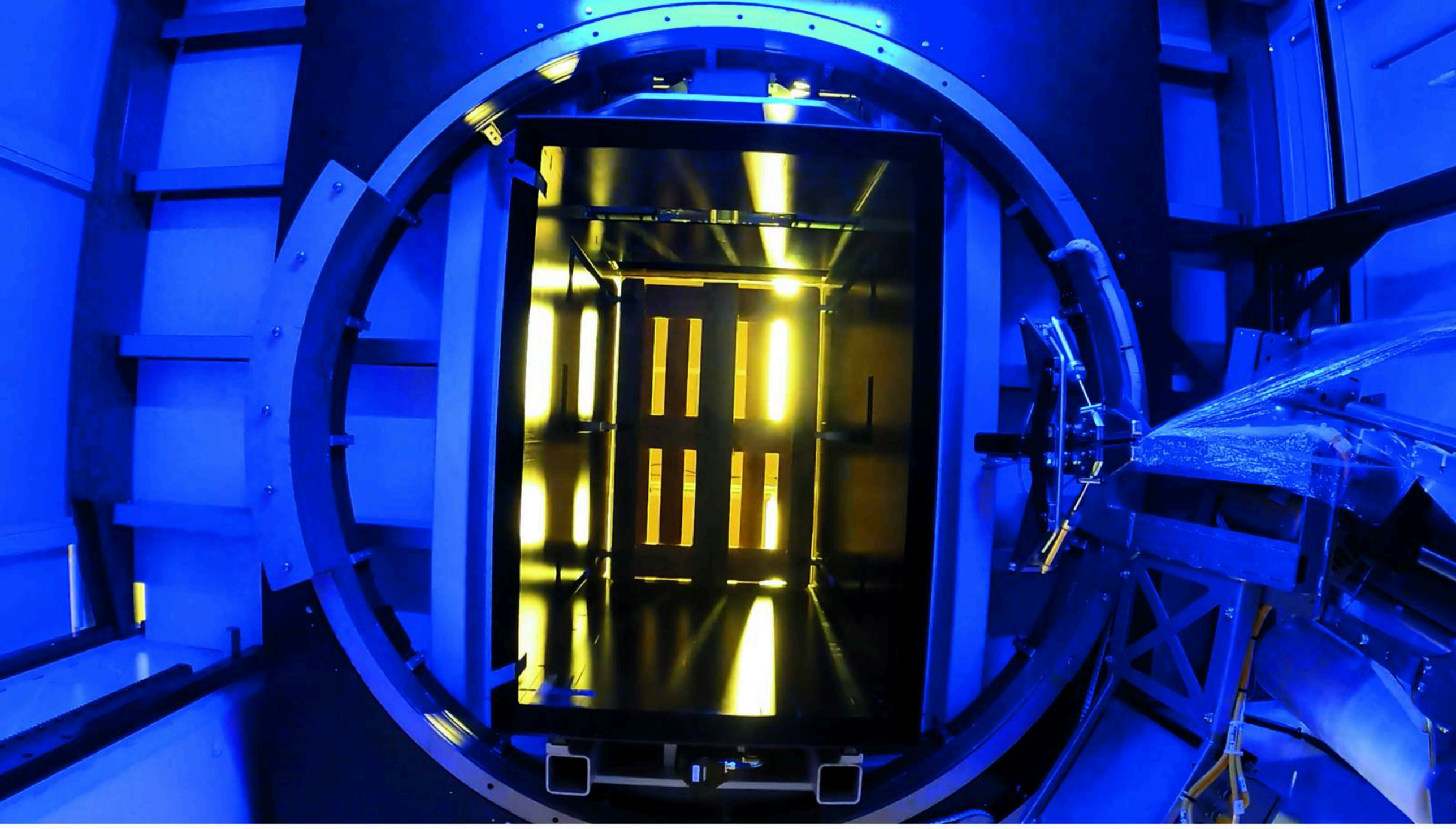


The Solution

At Photoneo, we are proud to collaborate with Yaskawa to deliver groundbreaking solutions, revolutionizing the palletizing landscape.

By merging Photoneo's state-of-the-art 3D vision technology with Yaskawa's robotic ecosystem, we are setting new standards for automation in warehouses and distribution centers. But how did the entire journey materialize? It all starts with PackMaster.



The Solution

In collaboration with Yaskawa Nordic AB, WSR Solutions developed PackMaster, a flexible palletizing solution that fully embraces Photoneo MotionCam-3D's capability to scan in motion. Designed for businesses looking to enhance their palletizing processes, PackMaster offers a modular approach.

As demands increase, it can start with manual operation and seamlessly upgrade to a fully automated system.

Palletization

A key component enabling PackMaster's automation prowess is the integration of multiple Photoneo MotionCam-3D cameras. These advanced sensors provide the Yaskawa robots with real-time, high-precision 3D vision, the crucial element for effective mixed-case palletizing.

MotionCam-3D captures depth information, creating a detailed 3D model of the products on the conveyor, even in motion. This allows the robots to see the shape, size, and orientation of each item, regardless of its position or the complexity of the surrounding environment.

Products typically move along a conveyor belt, and stopping them for scanning would drastically reduce throughput. The MotionCam-3D's in-motion scanning capability allows the robots to identify and locate products on the fly, without interrupting the flow of goods.

The cameras capture accurate 3D data of the moving products, which is then processed by the system to generate the optimal palletizing strategy. This real-time data acquisition and processing enables the robots to pick and place items with exceptional precision, even with varying product sizes and shapes.

How It Works

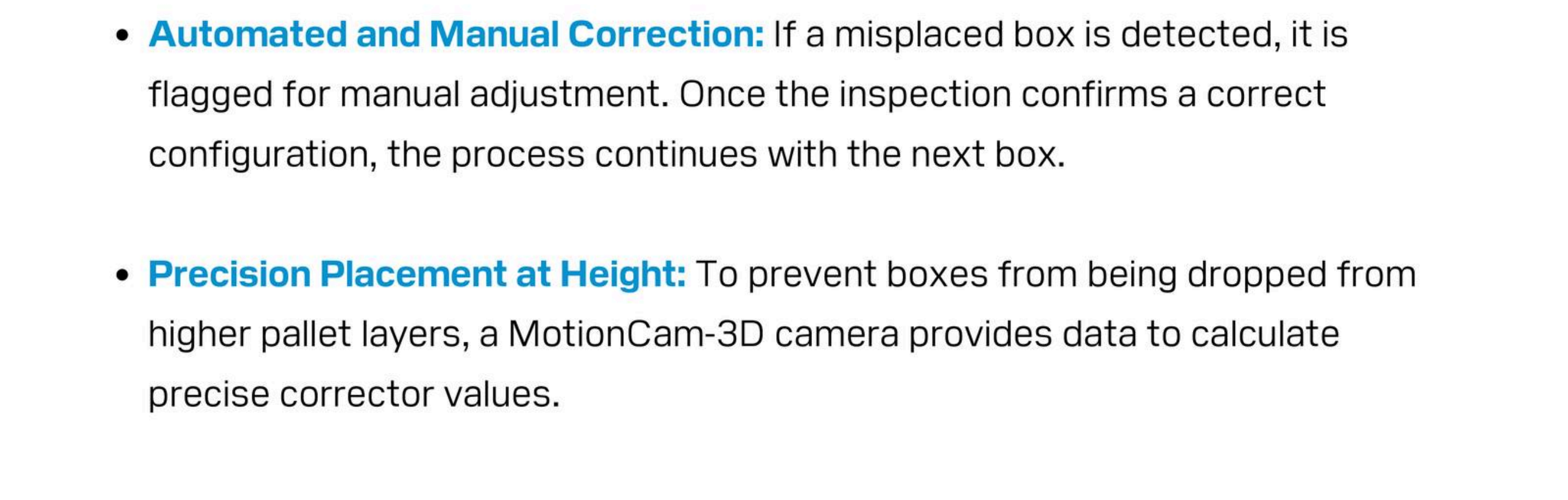
- **Inbound Verification:** The vision system first confirms that incoming boxes are correctly oriented and sized for palletizing. This ensures smooth downstream processing and prevents errors.

- **Dynamic Collision Avoidance:** Approved boxes are picked up by the robot, and Yaskawa‘s RapidPlan technology calculates a collision-free motion path in real time, thanks to MotionCam-3D’s in-motion scanning capacity.

- **Continuous Pallet Inspection:** After each box placement, a 3D camera scans the pallet. This scan is compared against the theoretical “built plan” to verify accurate placement.

- **Automated and Manual Correction:** If a misplaced box is detected, it is flagged for manual adjustment. Once the inspection confirms a correct configuration, the process continues with the next box.

- **Precision Placement at Height:** To prevent boxes from being dropped from higher pallet layers, a MotionCam-3D camera provides data to calculate precise corrector values.



Shared Vision, Smarter Future

Optimal Space Utilization

Improved Stability and Security

Increased Throughput and Efficiency

Quality Control and Traceability

Results And Benefits

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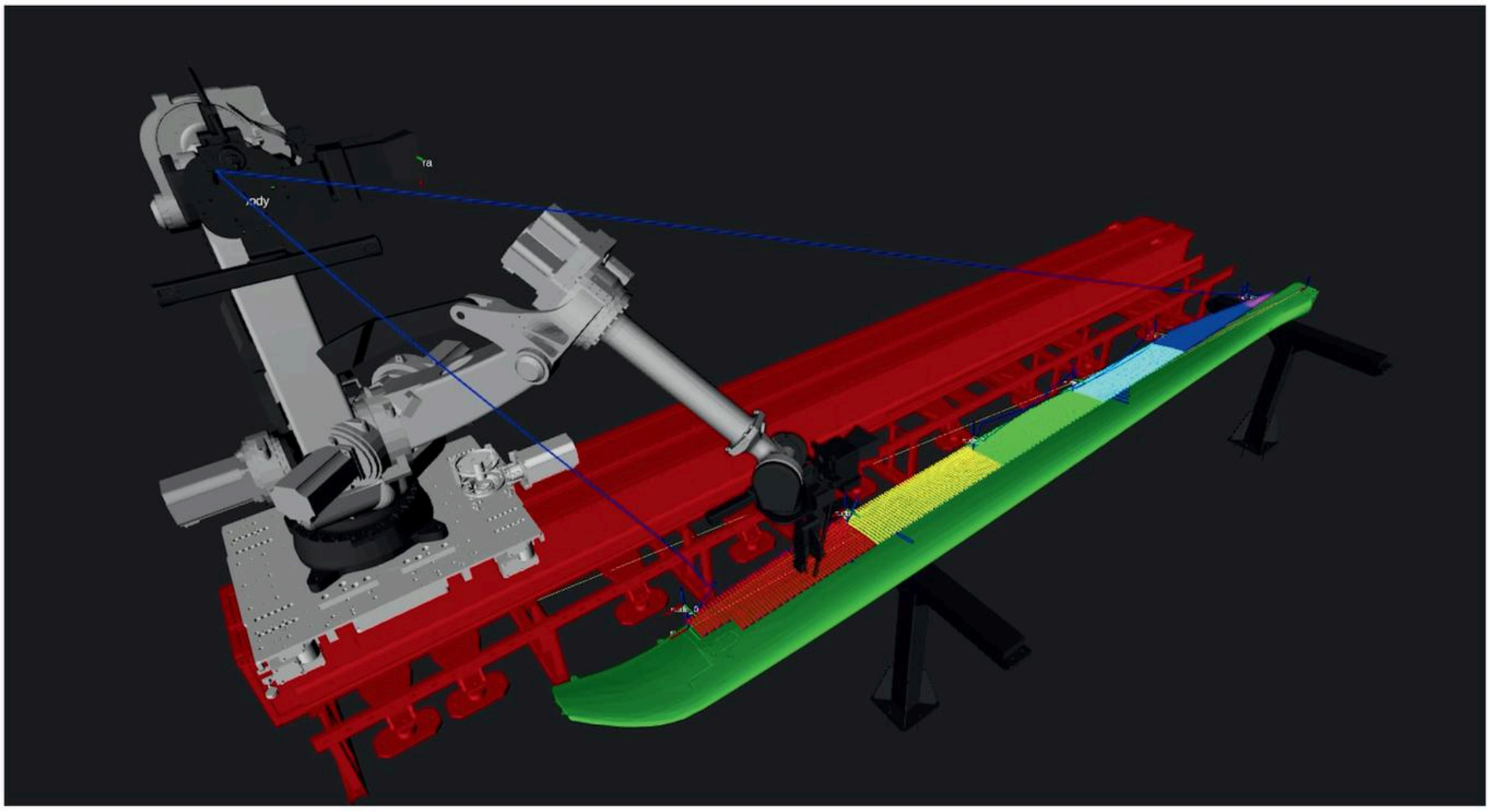
7-Meter Long Blades, Any Defect Can Be Fatal.

Can Machines Take Over?



USE CASE

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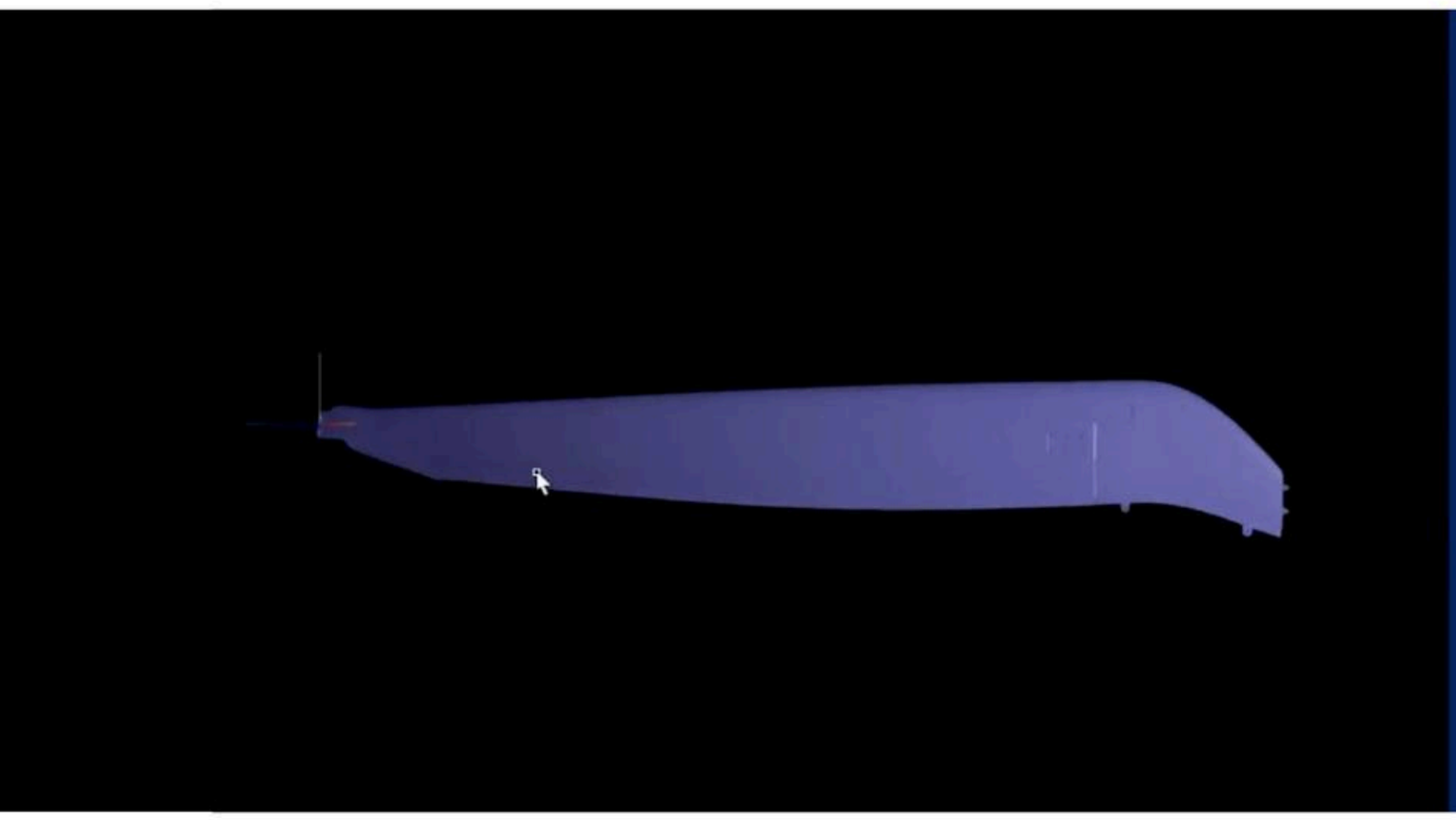
Overview

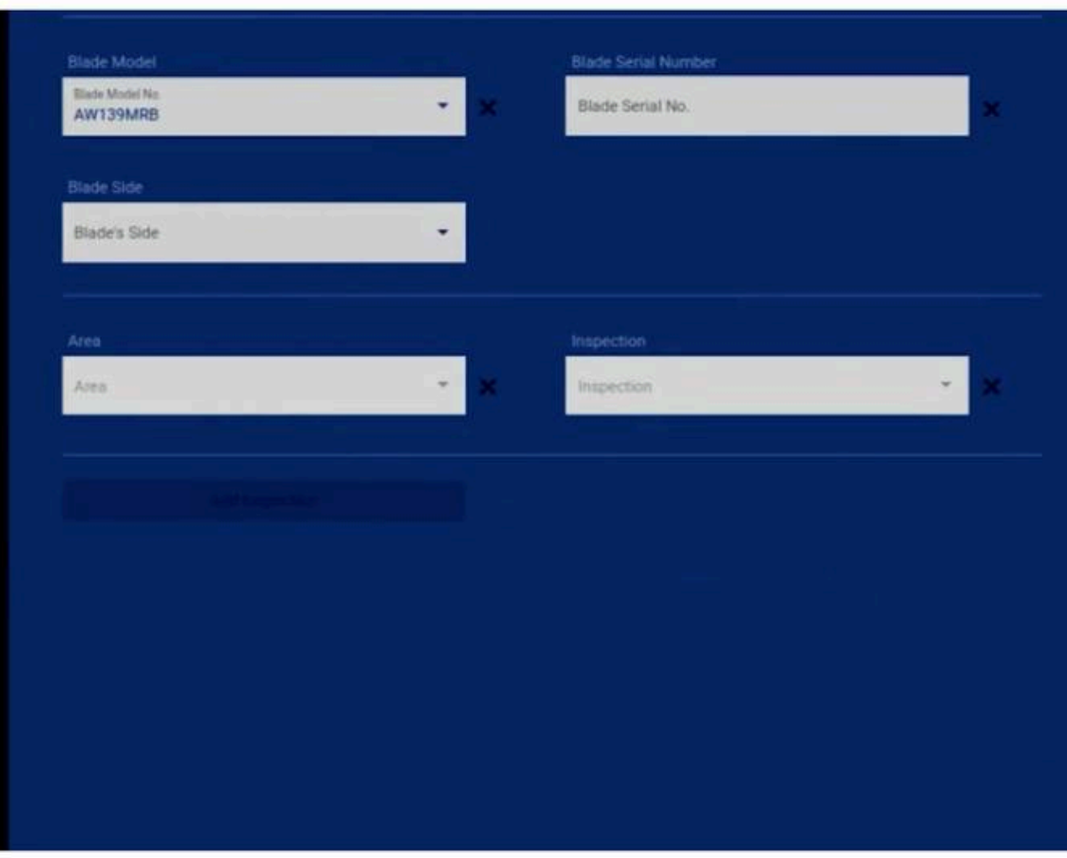
Meet Comau, our long-time partner and a titan in industrial automation and robotics. Founded in 1973 in Italy, Comau demonstrates a decades-long tradition of expert industrial automation and robotics.

Recently, a leading aerospace company challenged Comau with an exquisitely demanding task - that’s when the time came to join forces.

Spanning 1.5 years, the following project laid a new standard for the automated inspection of helicopter rotor blades.

Partner: Comau
Industry: Aircraft Manufacturing
Application: Vision Guided Robotics, Inspection
Technology: MotionCam-3D





The Challenge

Automating Helicopter Rotor Blade Inspection

Known for high standards, Comau's client faced challenges verifying helicopter rotor blades' integrity—traditional manual processes proved inefficient since rotor blade inspections require non-destructive testing (NDT) methods, which are labor-intensive, time-consuming, and rely heavily on the skill of experienced operators.

The company's requirements for helicopter rotor blade inspections included high accuracy, reliability, and scalability. To meet these demands, Comau sought to automate its rotor blade inspection and maintenance procedures.

But what's so tricky about inspecting helicopter blades?

Non Destructive Testing

Strict Inspection Processes

Helicopters are renowned for their speed and versatility, but they require stringent safety regulations, especially for critical components like rotor blades.

These blades generate immense force for flight, making even minor defects potentially catastrophic. Regular inspections are crucial to ensure safety and prevent costly repairs.

There’re 2 main challenges in the inspection process.

Helicopter Blade Inspection Challenges

Composite Materials: Many modern rotor blades use composite materials for their high strength-to-weight ratio. However, these materials are challenging to inspect effectively due to their thin layers and complex structures.

Complex Geometry: The intricate design of rotor blades, with their angled and curved surfaces, adds to the inspection difficulty. Traditional NDT equipment can struggle to access and analyze these complex shapes.

Objective

- **Ensuring accurate 3D acquisition of reflective rotor blade surfaces.**
- **Achieving precise pose estimation and registration for dynamic recalculation of robotic trajectories.**
- **Stitching rotor blade sub-sections into a comprehensive model.**
- **Meeting stringent accuracy and reliability standards in a challenging inspection environment.**

To ensure the safety and reliability of helicopters, a robotic solution capable of autonomously inspecting rotor blades is essential. This solution would revolutionize the inspection process, streamlining it and minimizing human intervention.

The robot would be designed to meticulously examine rotor blades up to 7 meters in length, combining a visual inspection with tactile feedback from tapping tests. This comprehensive approach would allow for a granular analysis of the blade's structural integrity, exceeding thousands of data points.

By automating the inspection process, the robot would significantly reduce inspection times and minimize operator dependency.

The Solution

Automated Robotic Inspection solution

3D Acquisition and Stitching

We provided PhoXi 3D Scanner L, mounted on a robotic arm to capture high-resolution 3D images of rotor blade sub-sections. These images were stitched together to reconstruct the entire rotor blade surface.

Pose Estimation

The reconstructed rotor blade scene underwent pose estimation and registration to determine its precise position. This enabled recalculation of the robot's trajectories to perform the required inspections.

Inspection Execution

The robotic system conducted visual inspections and tapping tests, ensuring comprehensive coverage of the rotor blade surface with unprecedented granularity.

Project Outcome

Reducing inspection time significantly.

Enhancing accuracy and reliability.

Opportunities to upskill operators

Digitizing inspection processes and outcomes for better traceability and analysis.

Results And Benefits

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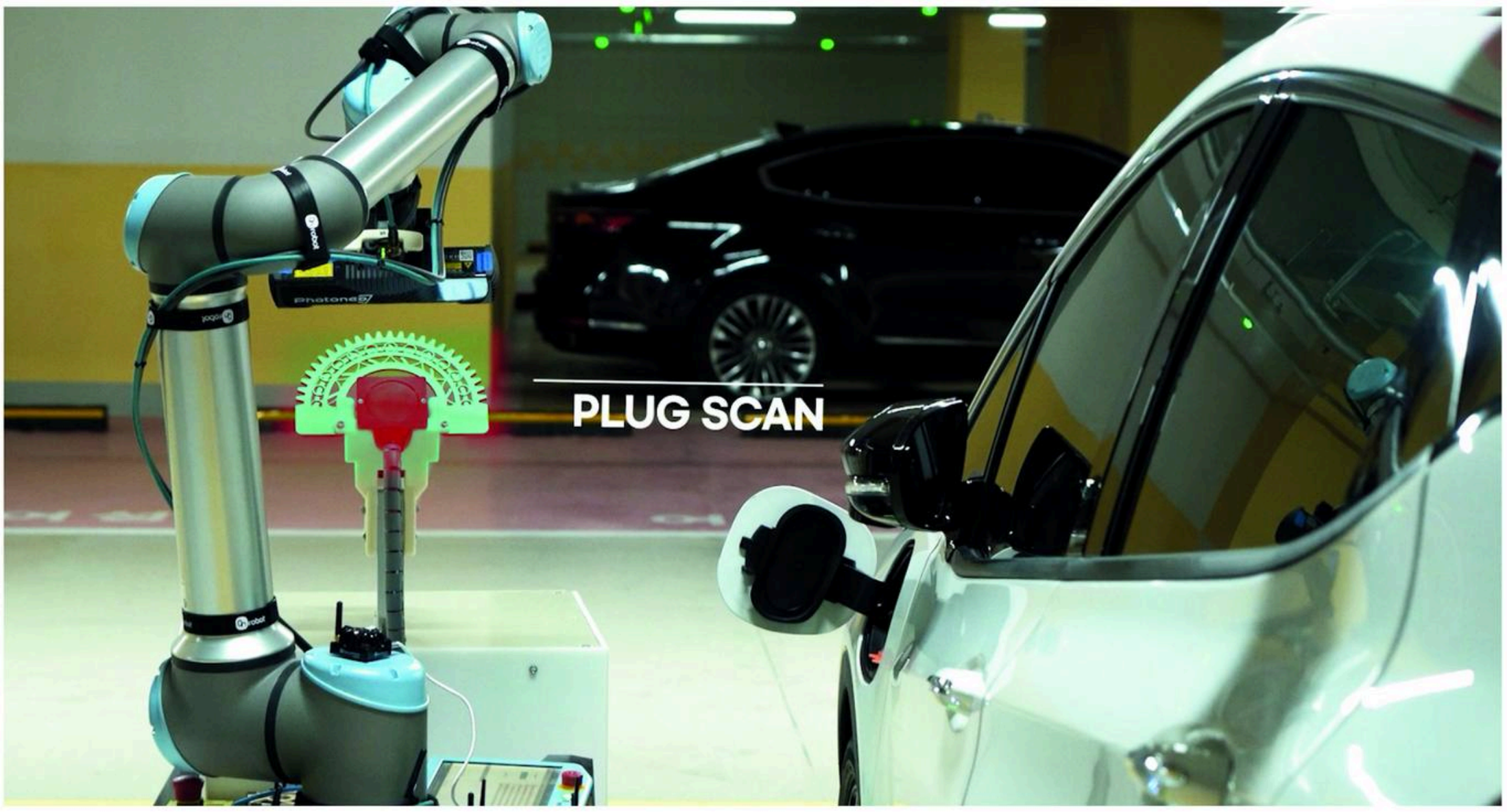
Vision-Powered EV Charging In A Button Click

Autonomous EV Fast Charging



USE CASE

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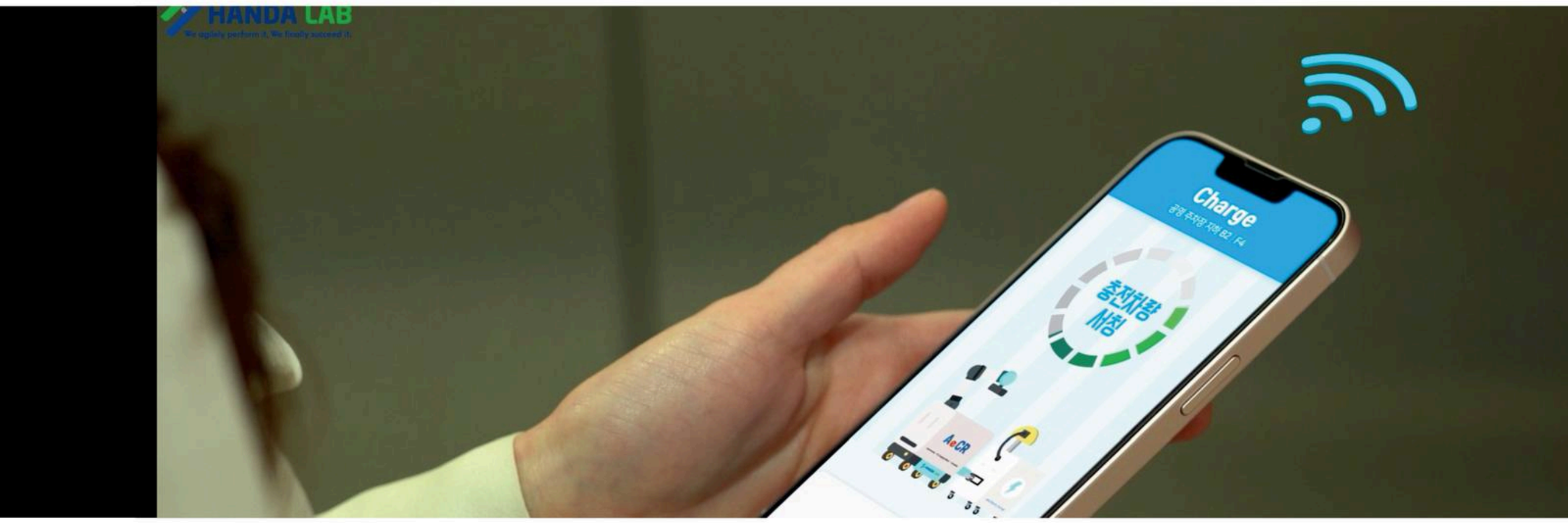
Overview

Densely populated urban areas, long queues, cluttered parking garages - all pose a significant challenge on charging your electric vehicle. Especially with the rise in demand, a new opportunity arose to automate the entire process. That’s why we partnered with Handalab to automate the process - from localization and charging to in-app payment.

By combining autonomous robots, 3D vision guidance, and intelligent software, together with Handalab, we're delivering cutting-edge VGR solution that makes charging your EV effortless and convenient. In essence, the resulting solution is not only about convenience but foremost caters to better accessibility to people with limited mobility.

Partner: **Handalab**
Industry: **Automotive**
Application: **Autonomous EV Charging**
Technology: **MotionCam-3D**



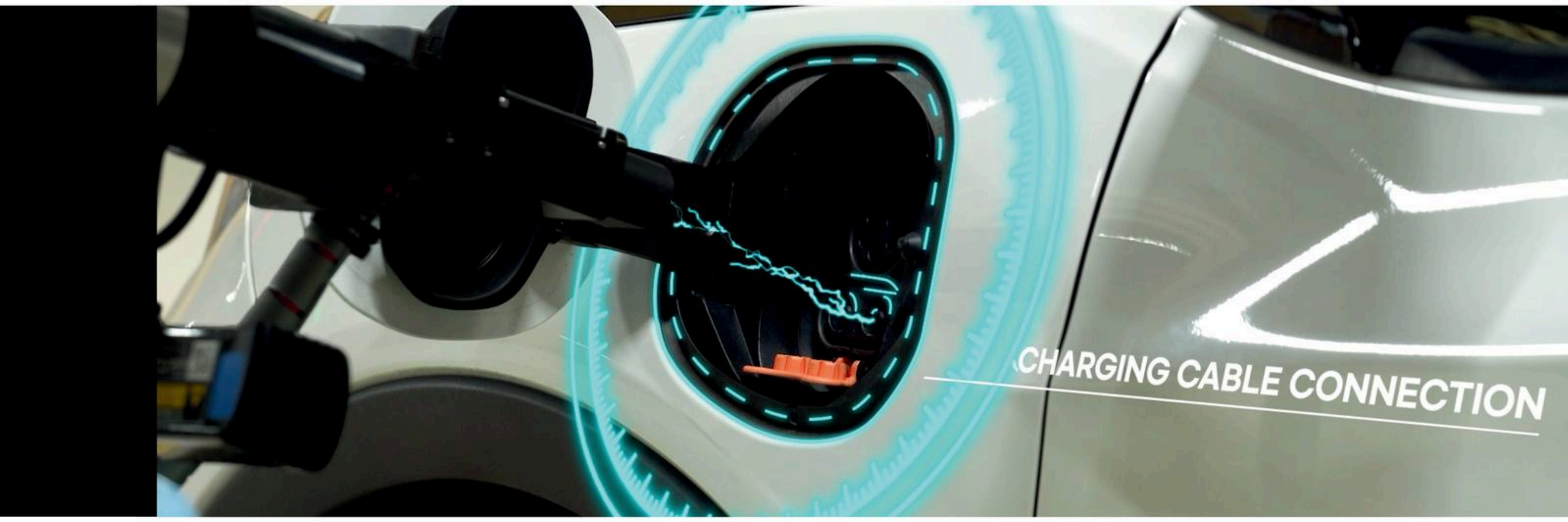
About Partner

Meet our partner, Handalab, a South Korean R&D powerhouse pushing the boundaries of AI vision and robotics since 2021. They're not just researching; they're developing cutting-edge solutions like deep learning-based monitoring systems that analyze data with exceptional accuracy.

Moving beyond, Handalab empowers businesses with smart data analysis for innovation and smarter decisions.

Notably, Handalab is also working on autonomous driving technology for robots, allowing them to navigate tricky environments independently.

And they're developing 3D-based intelligent robot control solutions, which means robots that can understand and interact with the world around them more effectively - and it’s this vertical that brings Handalab ahead of a challenge we were honored to solve together.



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The Challenge

Experience EV charging at the touch of a button.

The increasing popularity of electric vehicles (EVs) has led to a growing demand for accessible and convenient charging infrastructure. Traditional charging stations can be inconvenient due to limited availability, long queues, and the need for drivers to park precisely to access charging ports.

This challenge is particularly acute in densely populated urban areas with limited space for dedicated charging infrastructure. Especially individuals with mobility challenges may find traditional charging methods difficult.

Peak Accuracy For Autonomous Robots

To meet the rising demand for electric vehicle (EV) charging solutions, Handalab has engineered an innovative autonomous EV charging robot. This technology combines a mobile rapid charger with a sophisticated robotic system capable of autonomously navigating and docking with EVs.

Crucially, this robot utilizes a Photoneo 3D vision system to ensure pinpoint accuracy when connecting to an EV's charging port. This advanced technology allows the robot to identify and precisely connect with the charging port, even in challenging environments like crowded parking garages.

The Solution

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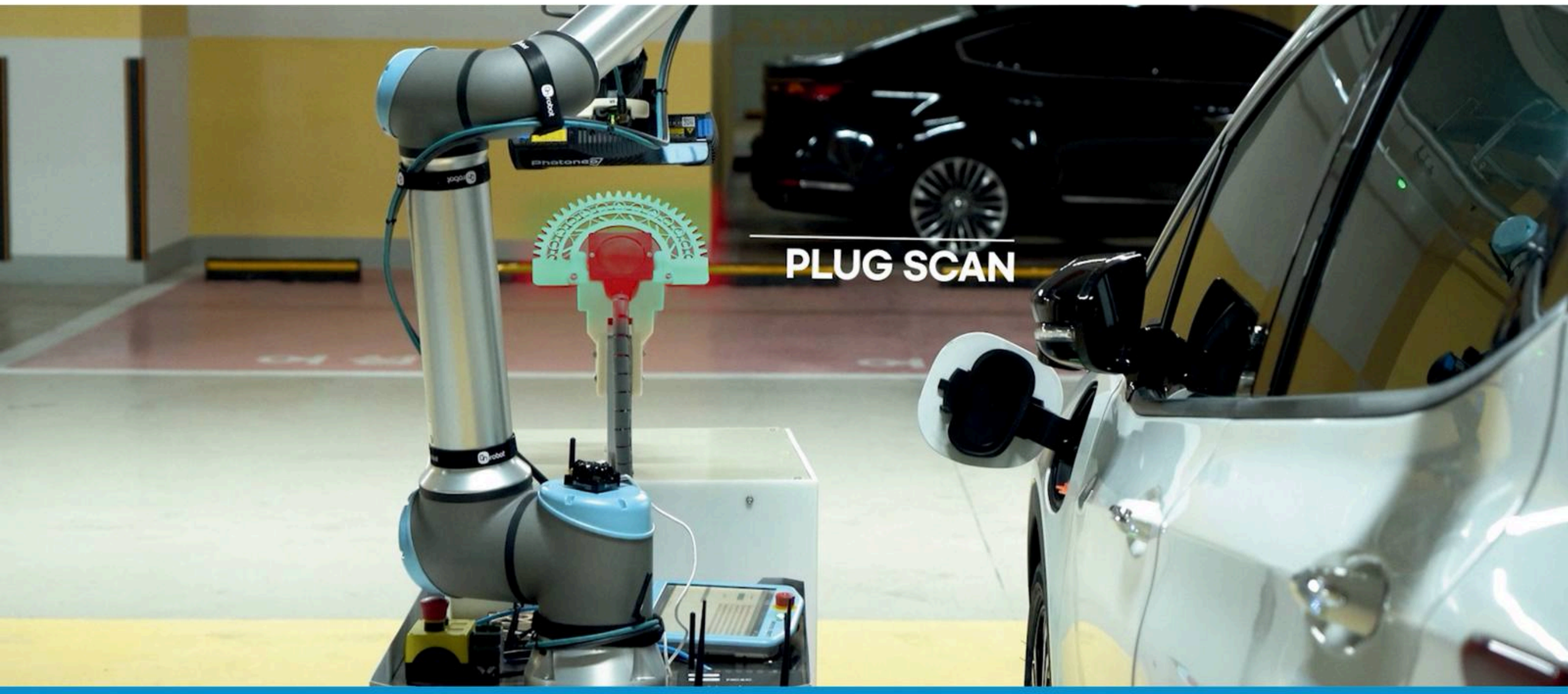
The Solution

- **Autonomous Driving Robot:** Equipped with advanced sensors and AI-powered navigation, the robot can autonomously navigate parking lots to reach the designated EV.
- **Mobile Rapid Charger:** A pre-charged mobile unit that provides efficient and fast charging, minimizing downtime for EV owners.
- **Control System:** A sophisticated system that manages the robot's navigation, charging process, and communication with the user app.
- **Photoneo's MotionCam-3D:** A 3D camera that accurately locates the charging port and connects the charging cable.
- **Mobile App:** Allows users to request charging services with a single click, track the robot's location, and monitor the charging progress.

The Innovation

- **Convenient Charging Service:** Users can request charging through the mobile app, eliminating the need to search for available charging stations.
- **Space-Saving:** The robot requires minimal space, utilizing areas like disabled parking spots when not in use, making it ideal for crowded parking lots.
- **Efficient Rapid Charging:** The mobile rapid charger, pre-charged during off-peak hours, ensures fast and efficient charging, minimizing waiting times.
- **Accurate Docking:** Utilizing Photoneo 3D camera technology, the robot accurately identifies and connects to the EV's charging port, ensuring reliable and safe charging.
- **3D Scanning in Motion:** Thanks to MotionCam-3D's ability to scan dynamic scenes in motion, the EV charging robot can see the scene in real time.



Jumping In

PEEK AT HOW IT WORKS

- Request a charge using the Handalab app from your parking spot.
- Handalab dispatches the nearest pre-charged autonomous charging robot.
- Equipped with advanced Photoneo robotic vision and AI-powered navigation, the robot effortlessly navigates the parking garage, avoiding obstacles and people.
- The Photoneo 3D vision system precisely locates your EV's charging port.
- The robot connects the cable and begins rapid charging.
- You monitor progress and process payment seamlessly via the app.
- Once complete, the robot disconnects and moves on.

Shared Vision, Smarter Future

- Space Optimization
- Increased Accessibility
- Enhanced User Experience
- Improved Convenience

Results And Benefits

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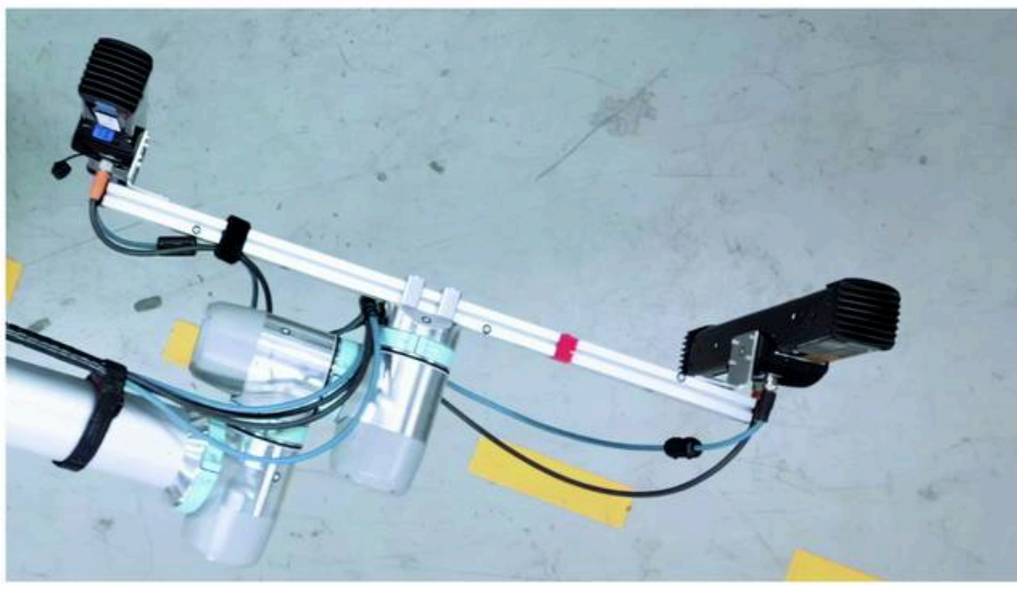
Watch the video

Increasing Car Safety With 3D Scanning In Motion



Automating Crash Car Inspection with 3D Cameras

USE CASE

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Overview

Speed, data quality, efficiency - discover how color 3D in-motion scanning enhances crash-car inspection.

Crash testing is crucial for automotive safety, but the process is expensive and time-consuming. Traditionally, after a crash test, technicians would manually scan the damaged vehicle using 3D cameras.

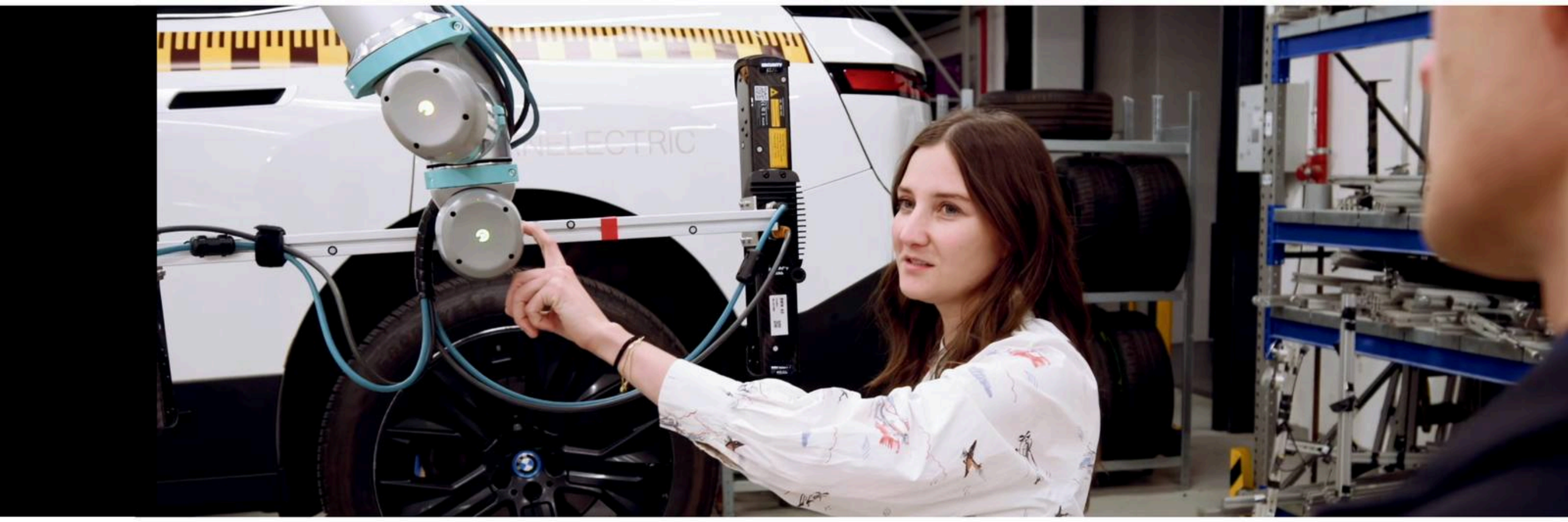
This process was slow, labor-intensive, and prone to inconsistencies.

Client: **Duwe-3d, RobCo, Evocortex**

Industry: **Automotive**

Application: **Crash test inspection**

Technology: **MotionCam-3D**



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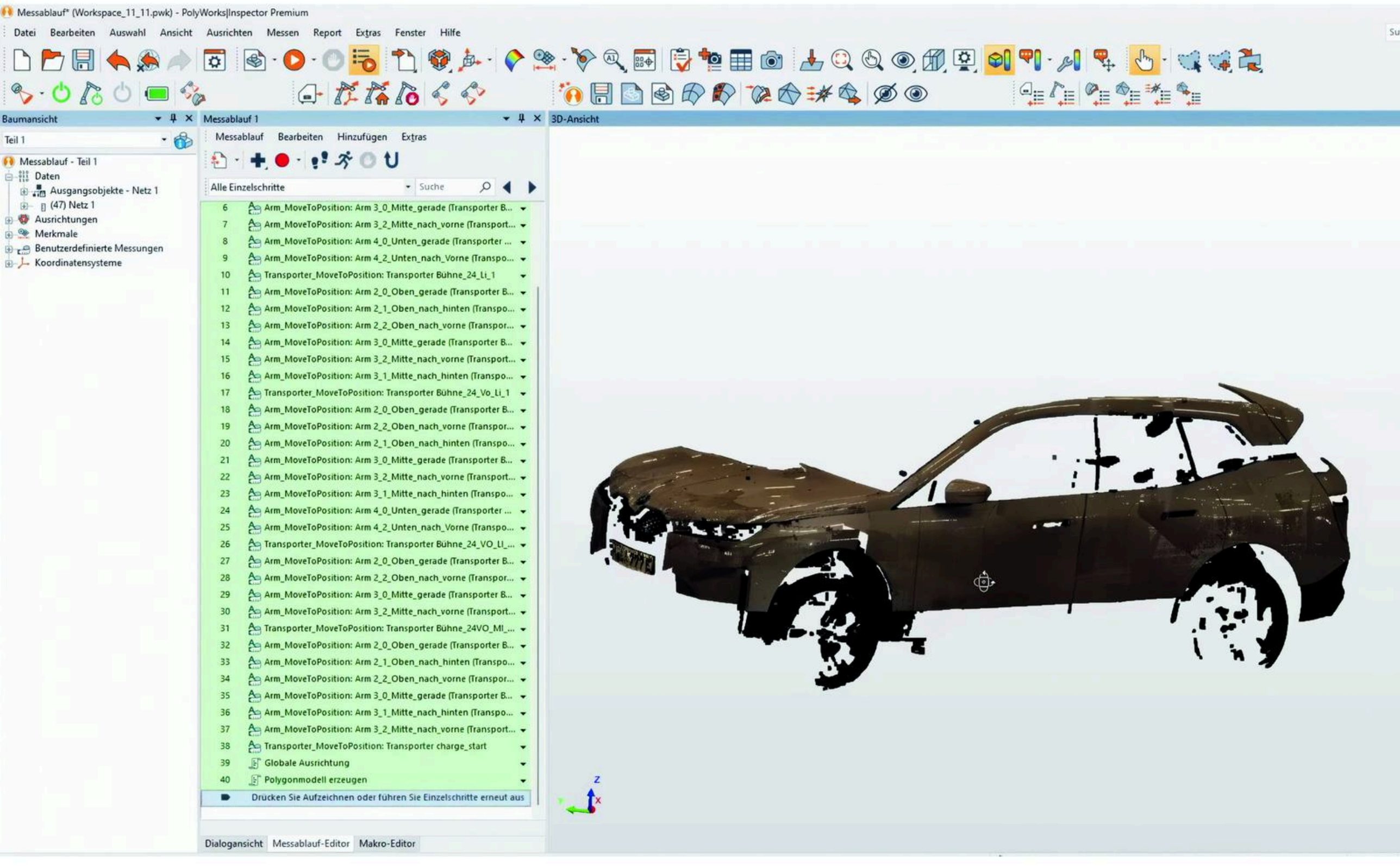
The Challenge

Manual Inspection Bottlenecks

A renowned German car manufacturer sought to automate this process to increase throughput, reduce costs, and improve the accuracy of data acquisition for comparing simulated crash models with real-world results.

The challenge was compounded by the unique nature of each crash test, no two crashes are identical, requiring a flexible and adaptable scanning solution.

Furthermore, the crash test environment itself presents obstacles, including equipment, debris, and varying car positions, making autonomous navigation and scanning complex.



An Integrated Inspection System

Duwe-3d, specializing in 3D metrology software PolyWorks, took on the challenge and developed RoboScan, an integrated system combining several key technologies.

The Solution

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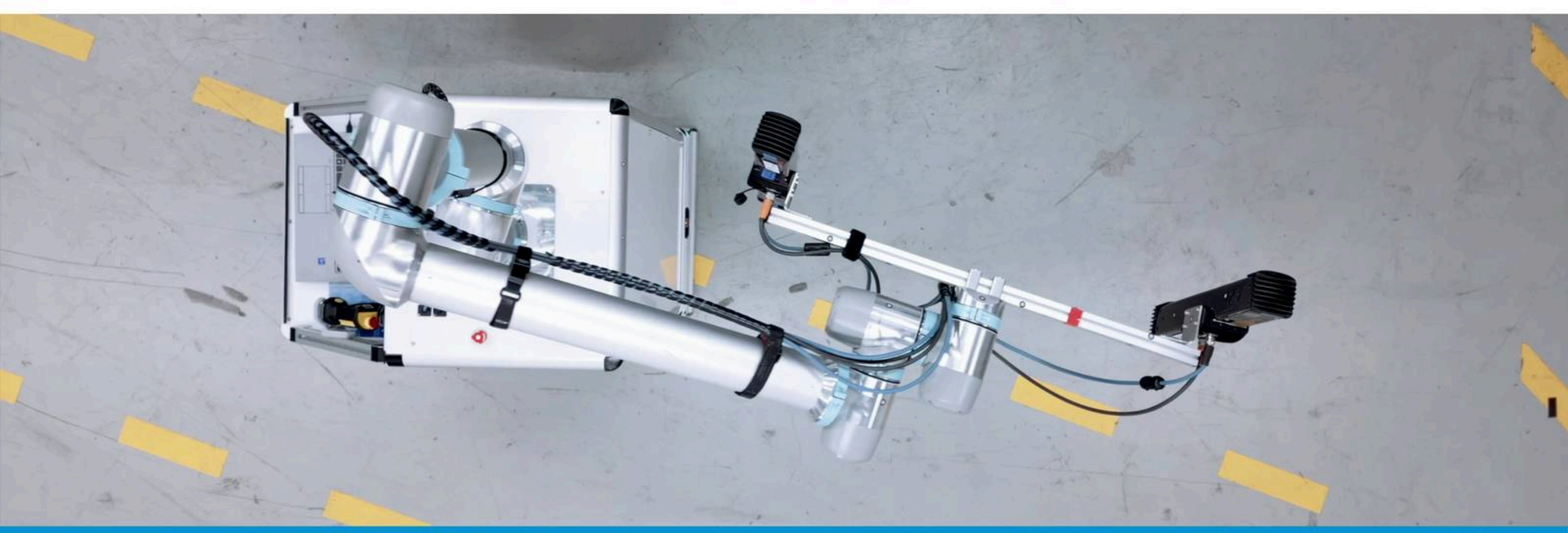
How It Works

Photoneo MotionCam-3D

Two Photoneo MotionCam-3D units form the core of the system, chosen for their speed, accuracy, and high-quality point cloud data. Using two 3D cameras simultaneously allowed for a wider field of view and faster data acquisition, significantly reducing scan time.

A crucial element was the integration of the Photoneo API, enabling seamless communication with the 3D cameras and allowing for precise control and data retrieval within the PolyWorks environment. MotionCam-3D units capture both point cloud and polygonal mesh data, along with color information, providing a comprehensive digital representation of the crashed vehicle.

Furthermore, MotionCam-3D's large scanning volume and high-speed capture enable synchronized area scanning, even for large objects. This creates accurate, blur-free 3D models for precise quality control, measurement, and digital twinning. Compatible with Photoneo 3D Meshing and other software, it boosts efficiency in high-fidelity 3D reconstruction applications, from automation to large-scale inspections.



How It Works

RobCo Cobot Arm

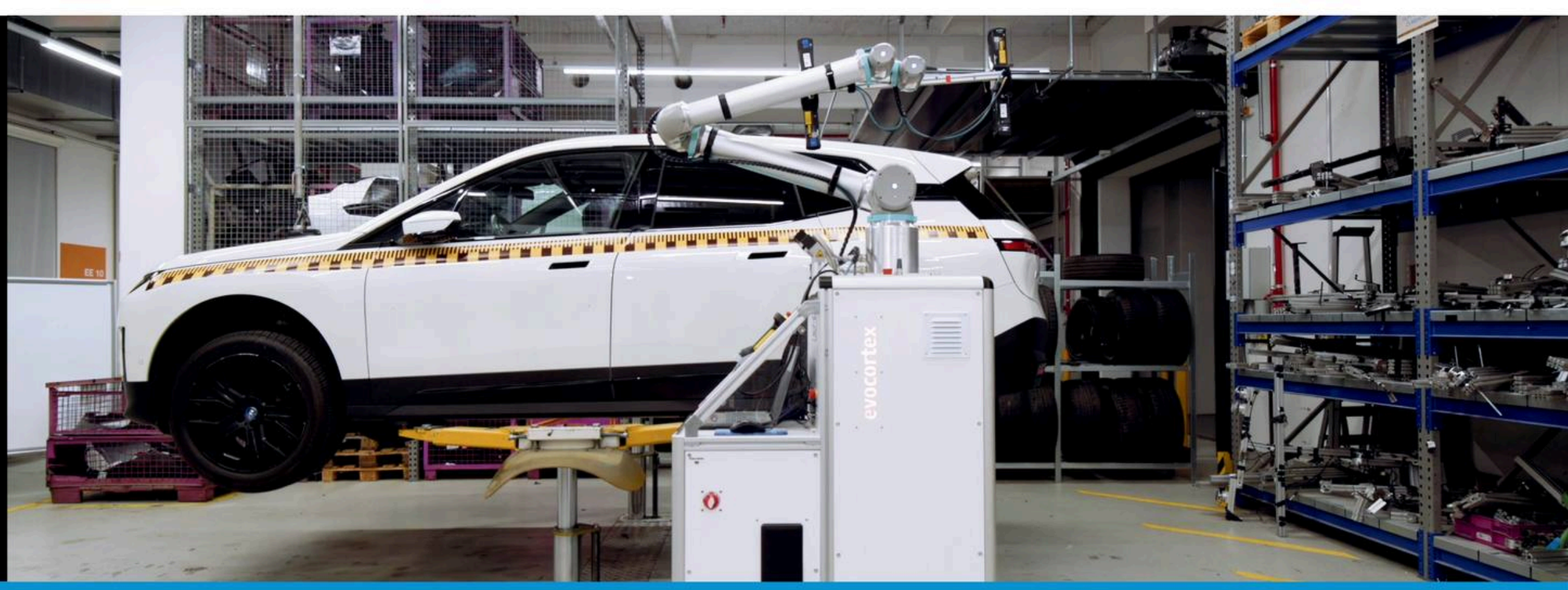
Photoneo cameras are mounted on a flexible RobCo cobot, enabling multi-angle scanning (including above and below the vehicle). Its collaborative design ensures safe operation alongside humans in shared workspaces.

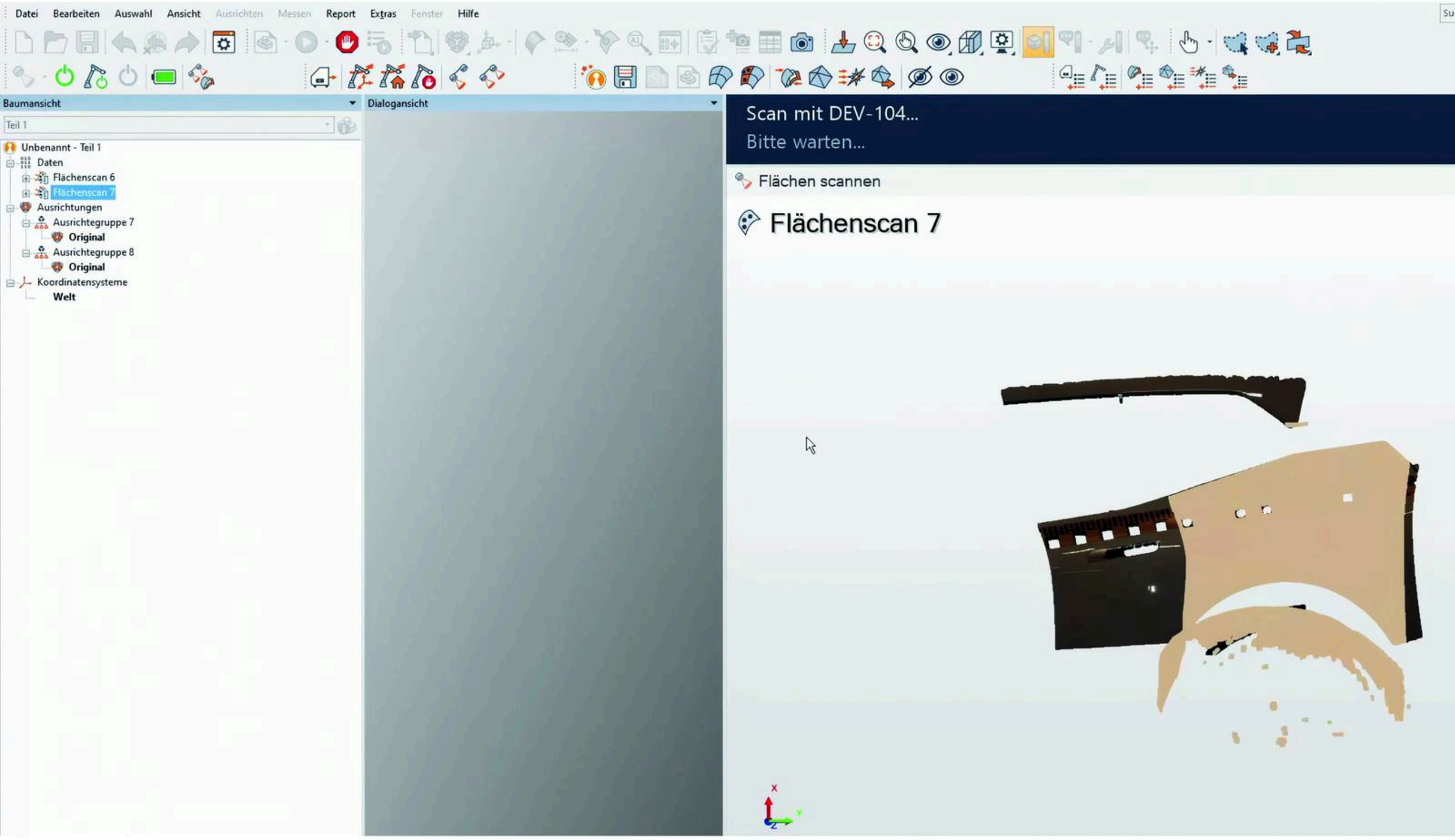
EvoCortex Mobile Platform

The system is integrated onto an autonomous EvoCortex platform, navigating the crash test area to reach pre-defined scanning positions for a complete 360-degree vehicle view.

PolyWorks Software

Serving as the central integration hub, PolyWorks utilizes Duwe-3d's D3D++ Pro Automation plug-in to manage hardware via various protocols. It synchronizes dual Photoneo MotionCam-3D units into a unified point cloud and compares scanned data against crash simulations to highlight discrepancies.





Setup and Scanning

1. Setup and Planning: The user defines a scanning sequence within PolyWorks, specifying the desired positions and orientations of the 3D cameras around the car. This sequence automates the data capture process, ensuring consistent and repeatable scans.

2. Autonomous Scanning: The RoboScan system, guided by the pre-defined sequence, autonomously navigates the crash test area. The EvoCortex platform moves the robot arm to the designated locations.

The RoboScan

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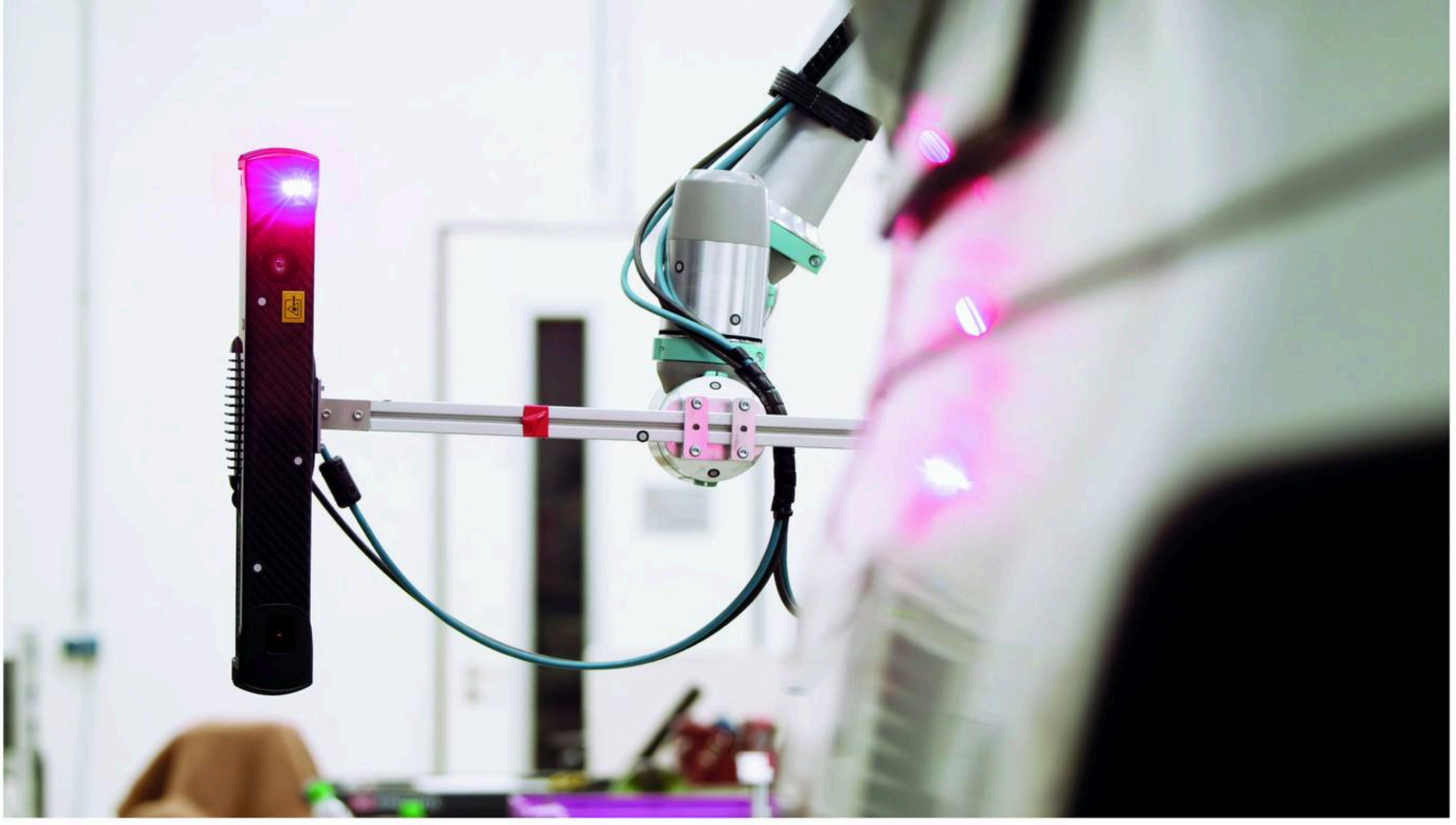

Data Processing

3. Synchronized Data Capture: At each position, the RobCo cobot arm precisely positions the two Photoneo 3D cameras. The D3D++ plugin triggers the 3D cameras simultaneously, capturing high-resolution 3D data and color information. The Photoneo API plays a critical role here, ensuring accurate and synchronized data acquisition.

4. Data Processing and Alignment: The captured data from the two Photoneo 3D cameras is automatically aligned within PolyWorks using the D3D++ plugin. This process creates a unified point cloud representing the entire crashed vehicle.

From Scan To Simulation Improvement

5. Comparison with Simulation: The aligned scan data is then compared to the original simulated crash model within PolyWorks. This comparison reveals discrepancies between the simulated and real-world crash behavior, providing valuable insights for refining the simulation models.

6. Simulation Improvement: The identified discrepancies inform engineers on areas where the simulation model needs improvement. By iterating on the simulation based on real-world crash data, the end client can improve the accuracy of their crash simulations and ultimately design safer vehicles.

Results
And Benefits

- **Increased Speed and Efficiency**
- **Improved Data Quality**
- **Seamless Integration**
- **More Detailed Insights**

Safer Cars, More Precise Inspection

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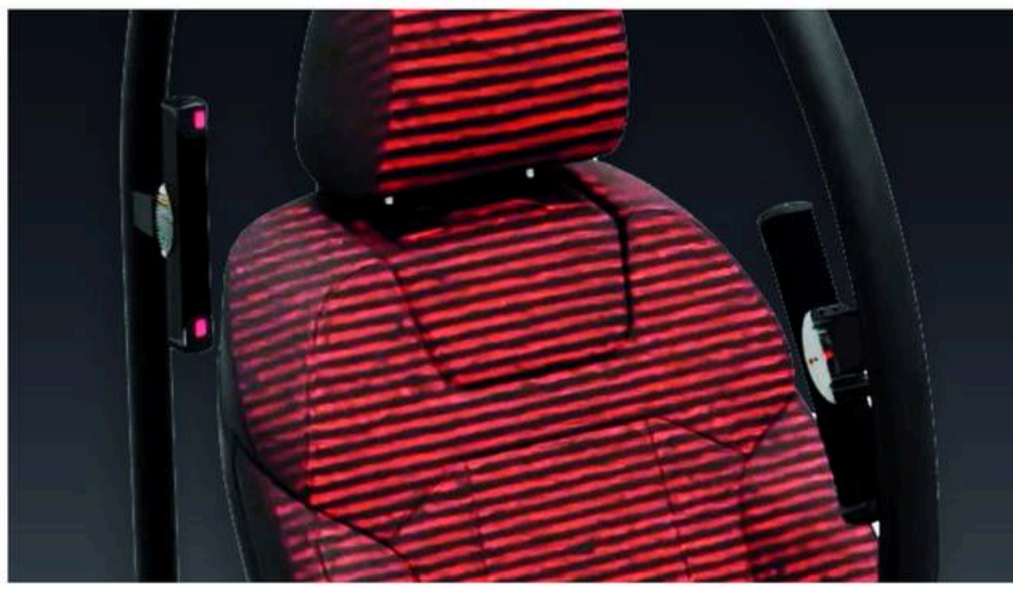
Synthetic Data Engine For Car Seat Inspection



100% Validated Inspection.
From The 1st Seat That Leaves The Line.

USE CASE

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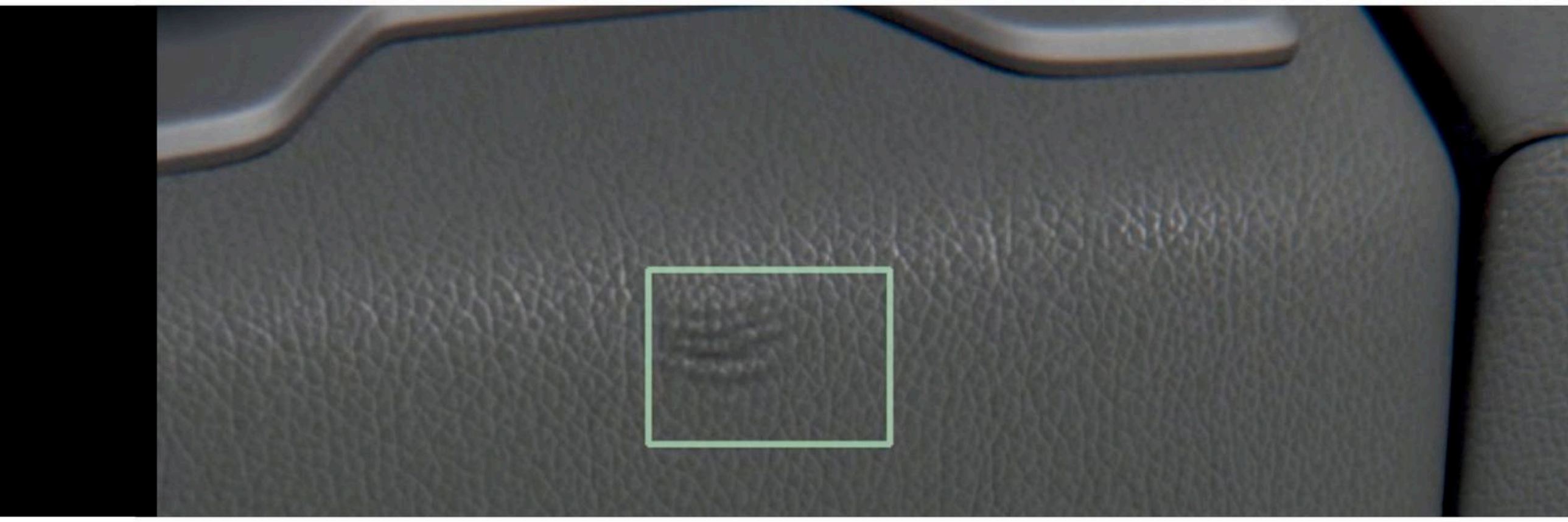
Overview

Even the 1st car seat that leaves the line has to be 100% inspected. With immense precision and speed. No CAD models.

How can we develop a rapid, adaptable, and cost-effective inspection solution for car seats? The solution must address challenges such as high product variety and frequent product changes.

100% inspection accuracy from the very first unit produced comes as a hard requirement.

Partner: ID Engineering
Industry: Automotive
Application: 3D Inspection
Technology: MotionCam-3D



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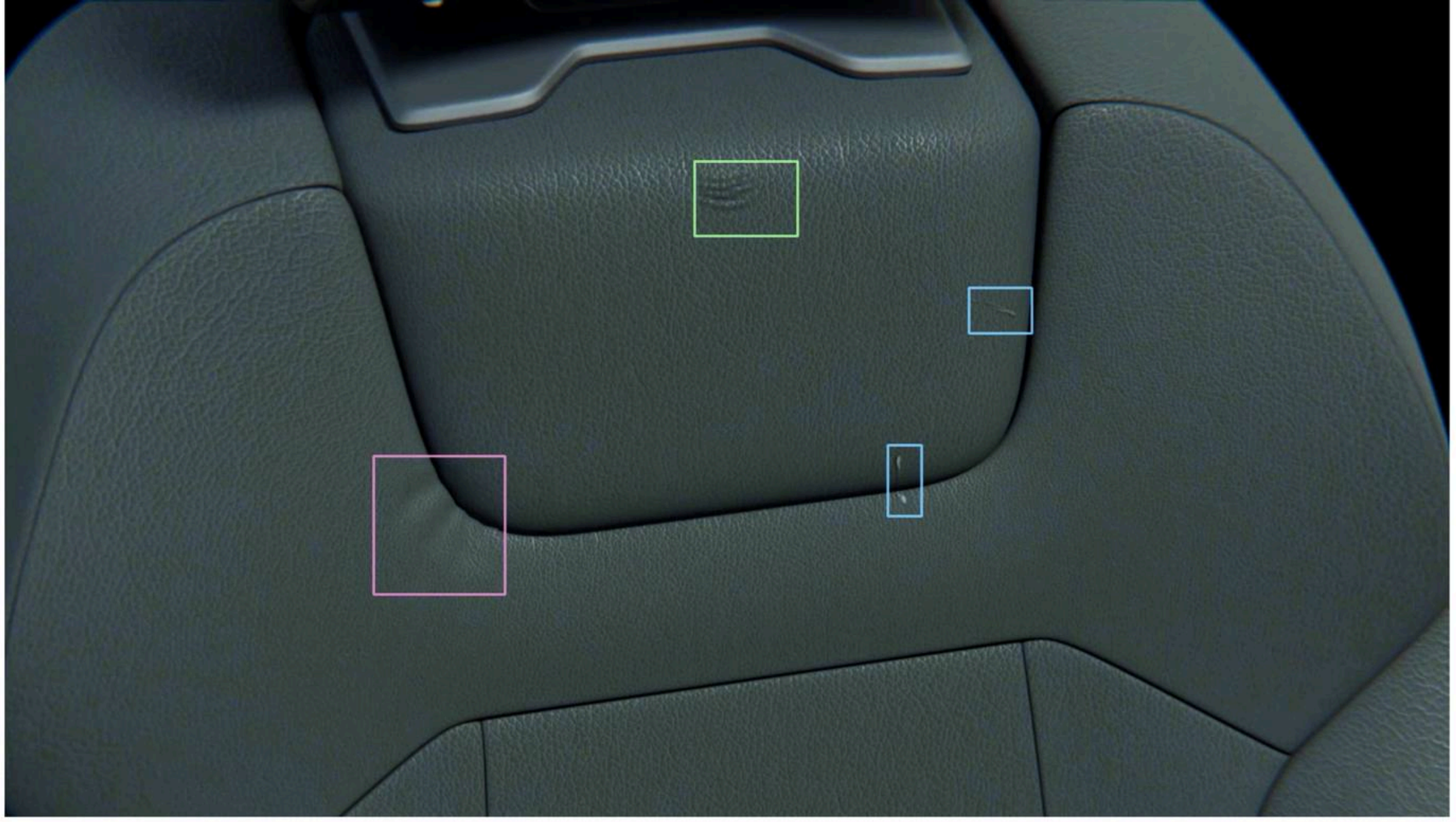

The Challenge

Quality Control vs. High Model Variety

To address car seat inspection challenges stemming from high model variety and frequent design changes, a leading automaker partnered with ID Engineering.

Current methods are insufficient: traditional imaging fails on low-contrast defects, and deep learning’s long training cycles cannot keep pace with rapid production.

These limitations create a quality control bottleneck that threatens both production throughput and overall product quality.

The solution required a complex system

100% Inspection from the First Unit - No training period is acceptable.

Rapid Adaptability: The solution must quickly adapt to new seat designs

Implementation Across Production Lines: The solution must be easily deployable

The solution must function independently of CAD model availability

Hard Requirements

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High-Fidelity Vision for AI-Powered Robotics

The project's initial shift to deep learning proved too slow for rapid development.

To meet requirements for 100% inspection from unit one—without CAD models and scalable via digital cloning—ID Engineering developed a hybrid solution.

This approach combines traditional image processing with deep learning classification, effectively eliminating the training data bottleneck.

The Solution

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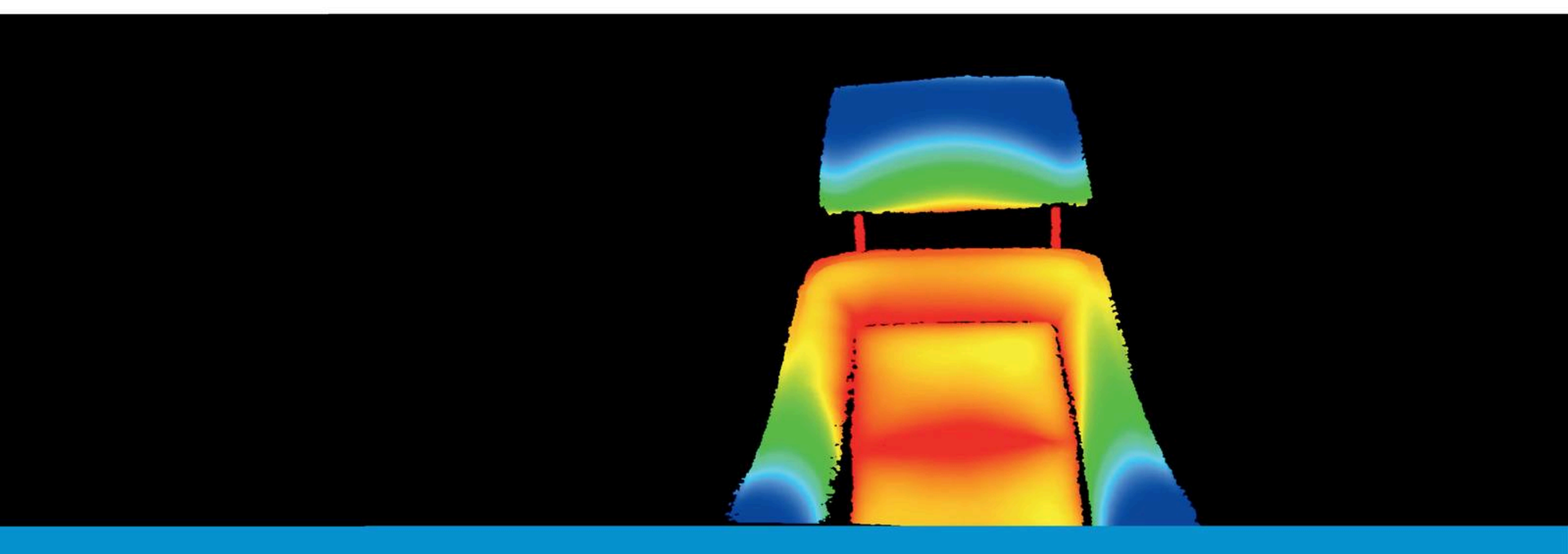
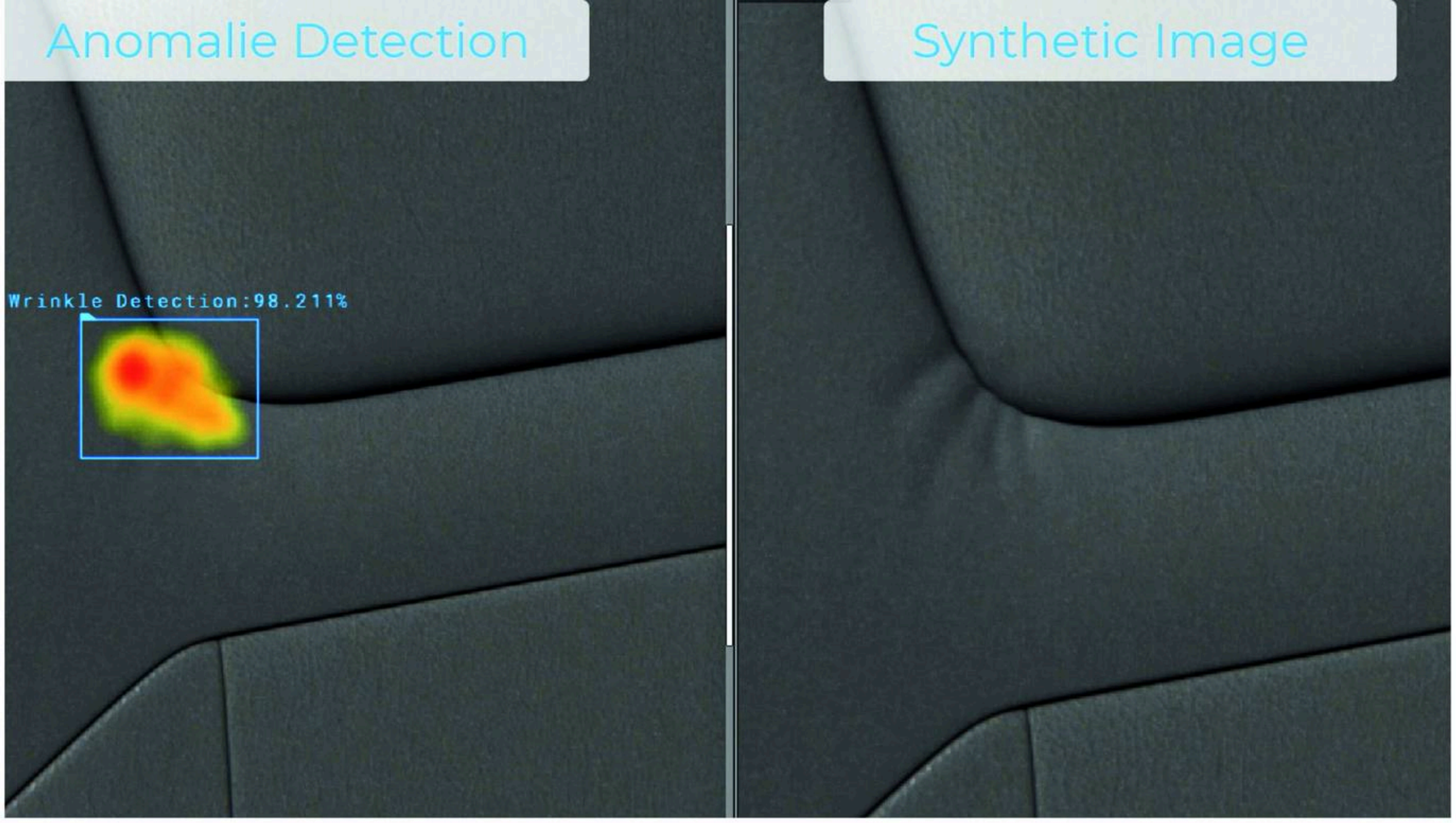
The Solution

- **Synthetic Data Generation:** When CAD 3D data is unavailable (a frequent occurrence with the diverse seat models), the MotionCam-3D's reconstruction capability is used to generate synthetic data. This synthetic data serves as the training ground for the deep learning network, eliminating the need for extensive real-world data collection for each seat model. This dramatically reduces the training time and allows for the inspection of new models virtually immediately. This addresses the crucial requirement of inspecting the very first unit produced.

- **Real-time 3D Feature Inspection**: Photoneo's MotionCam-3D is also used as a vision camera for inspecting seat features that require Z-axis height information. Crucially, the Photoneo system's ability to scan moving objects is essential here, enabling inline inspection on the production line. This allows for automated detection of defects that are difficult or impossible to identify with 2D imaging or traditional methods, directly addressing the low contrast defect issue.





How It Works

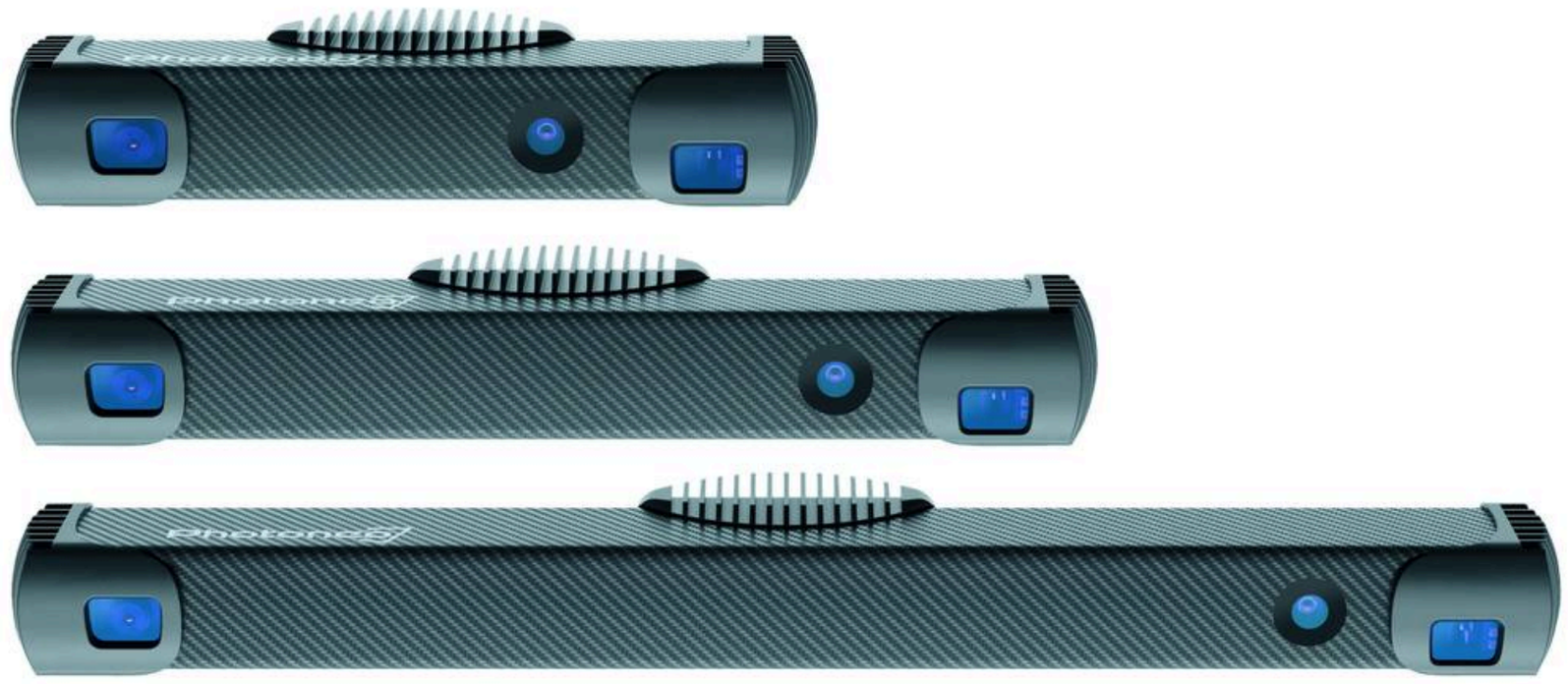
Project In Perspective

Traditional image processing techniques are first applied to the captured 2D images to enhance the low-contrast defects. This pre-processing step makes the defects more amenable to classification.

The deep learning model, trained using the synthetically generated data from the Photoneo scans, then classifies the pre-processed images, identifying and categorizing the defects.

This combination allows for both accurate defect detection and precise defect classification, addressing the customer's core needs.

The solution required a complex system



Rapid Implementation in just a few weeks.

No Real-World Data Needed: Uses synthetic data generation

Day-One 100% Inspection: Successfully detects low-contrast defects

CAD-Free Digital Cloning: Integrates seamlessly into production lines

Key Benefits

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Automating Car Body Production with Precise 3D Vision

Volkswagen Slovakia

USE CASE

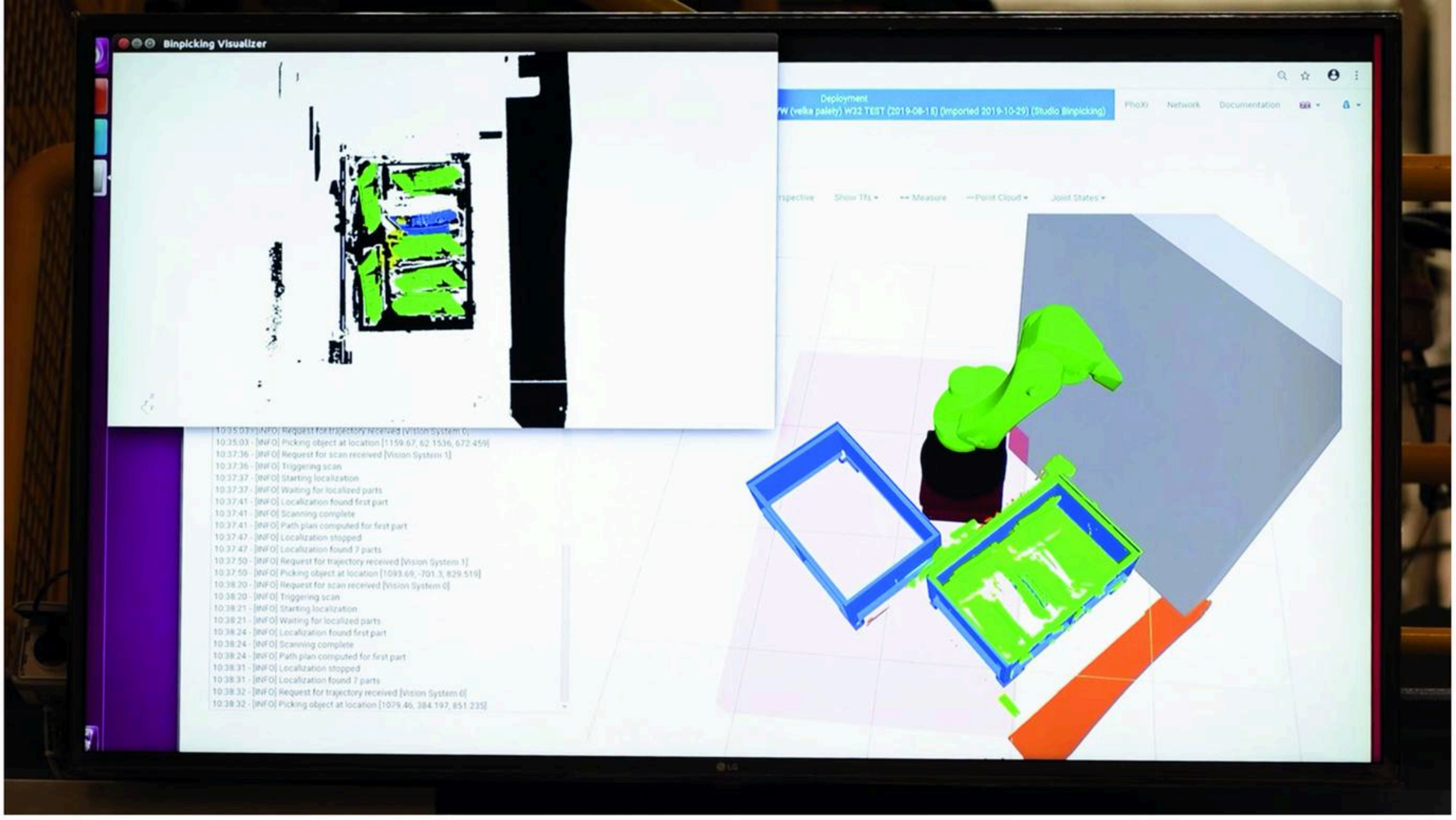
WWW.PHOTONEO.COM

Overview

Advanced Vision-Guided Robotics (VGR) is no longer a theoretical concept; it's a tangible reality transforming automotive manufacturing - there are numerous applications being deployed successfully in the automotive industry today.

Volkswagen Slovakia sought to automate a labor-intensive car body production process. Previously, components were haphazardly stored in floor-level containers, requiring manual retrieval and placement into a jig before robotic processing.

Client: Volkswagen Slovakia
Industry: Automotive Manufacturing
Application: Vision Guided Robotics, Machine Tending
Technology: PhoXi 3D Scanner



The Challenge

Manual step proved Inconsistent

- Labor-intensive manual process
- Inconsistent and inefficient manual retrieval
- Need for precise part identification and location
- Requirement for dynamic part handling
- Ensuring accurate placement into jigs
- Eliminating manual variability

High-Fidelity Vision for Robotics

Volkswagen aimed to eliminate these challenges by implementing a fully automated robotic system capable of direct part retrieval and jig insertion.

Photoneo, now part of Zebra Technologies, delivered a robust solution centered on high precision PhoXi 3D Scanner XL.

This scanner was critical in analyzing 3D CAD models of the components, accurately determining their precise position and orientation within the containers.

3D Vision Guidance with PhoXi 3D Scanner

- **Precisely Locate and Retrieve:** Identify and extract parts from the bin with pinpoint accuracy.
- **Dynamically Adjust Orientation:** Utilize a sophisticated gripper to adjust part orientation in real-time.
- **Accurately Place into Jig:** Consistently and reliably position parts into the jig, ensuring seamless integration with downstream processes.

The Solution

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Consistent Vision-Guided Operation

Enhanced Process Stability: Eliminated manual variability

Increased Repeatability: Achieved precise and repeatable part placemen

Reduced Production Costs: Minimized labor requirements

Increased throughput: The automated system was able to work constantly

Results And Benefits

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Automating Road Inspection: High-Speed 3D Scanning

In Any Environment

USE CASE

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Overview

Modern road inspection is moving from slow, subjective manual methods to automated 3D scanning at driving speeds.

Neousys' rugged Nuvo computers provide the "brain" for these systems, delivering industrial-grade performance in harsh, high-vibration vehicle environments.

By integrating Photoneo's MotionCam-3D as the precision "eyes," this partnership enables reliable, real-time pavement analysis on the open road.

Client: Neousys Technology
Industry: 3D Machine Vision
Application: Road Inspection
Technology: MotionCam-3D

Power Supply
(12-35V DC OUT)

Neousys SuperCAP-based
Power Backup Module

Neousys Nuvo Series Computer

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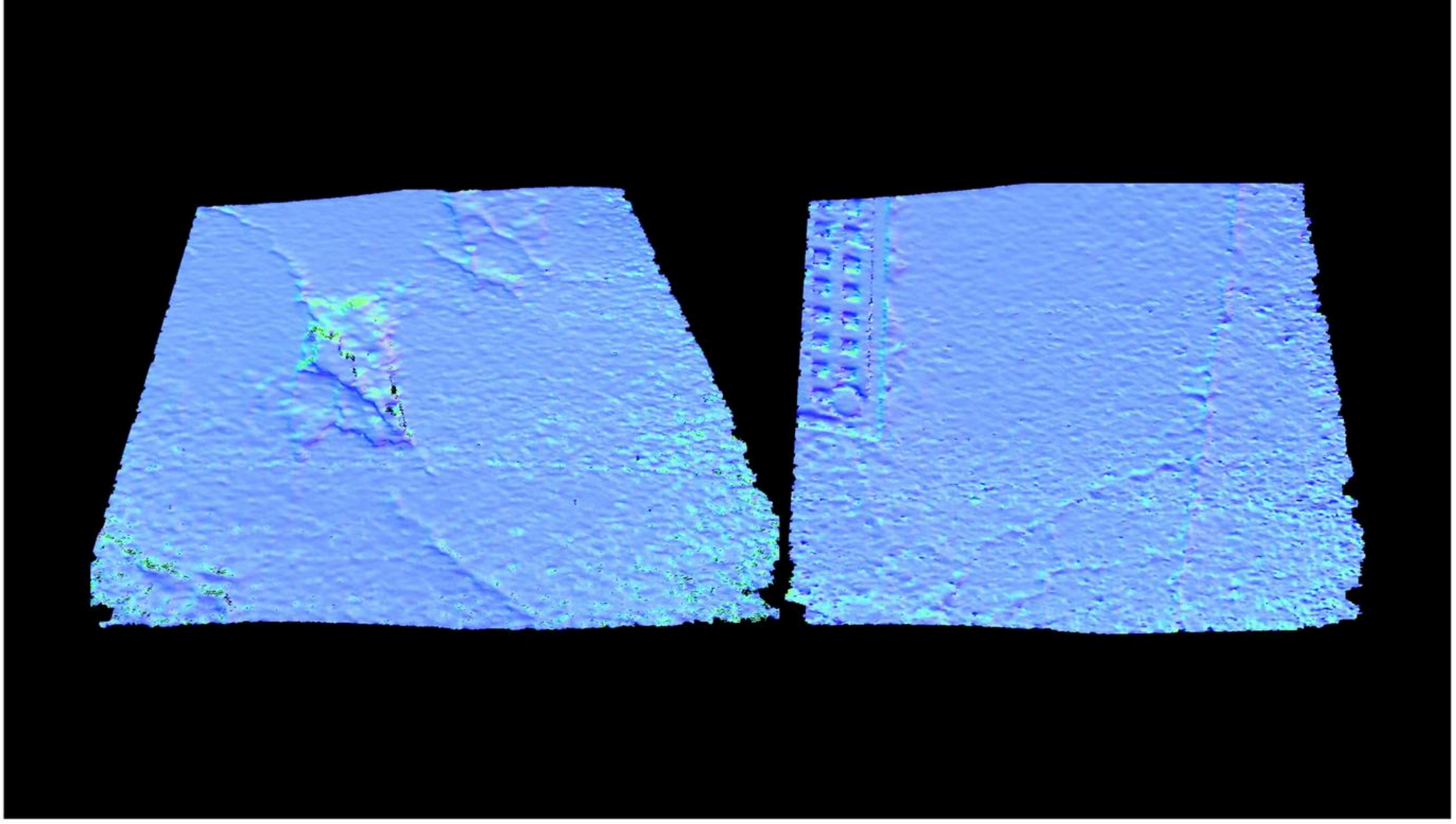


The Challenge

Taking High-Res 3D Data onto the Open Road

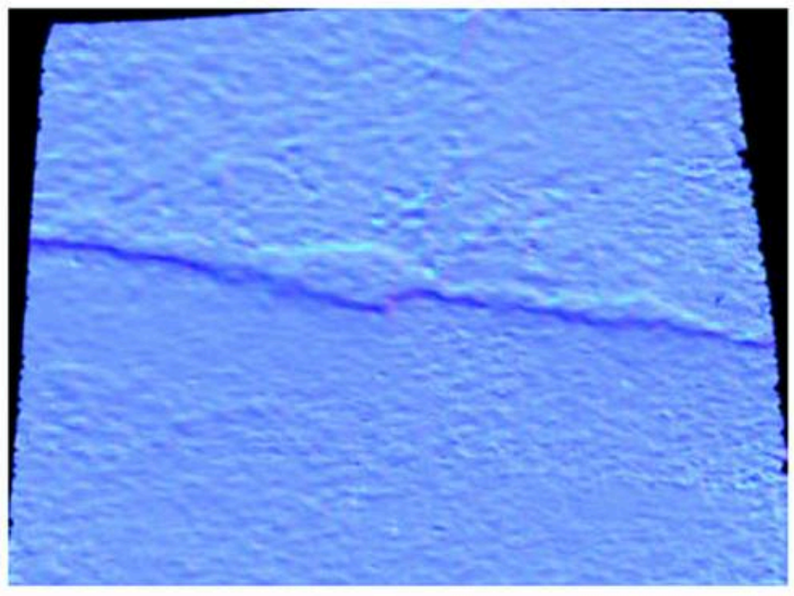
Our MotionCam-3D provides the perfect “eyes” for this application, but a complete solution requires a robust “brain.”

Capturing this much high-resolution 3D data at high speed generates a massive data stream that must be processed, fused, and stored in real-time. Deploying a high-performance computer onto a moving vehicle presents a severe set of engineering challenges:

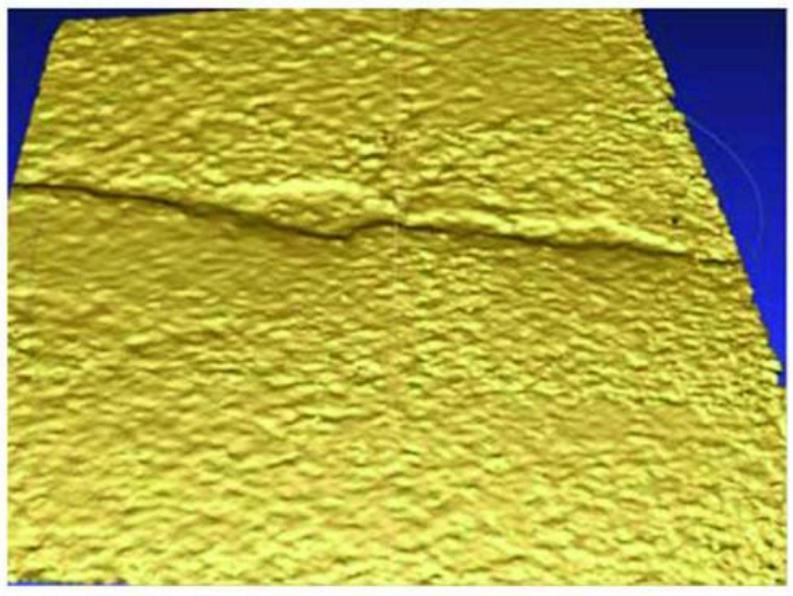
Key Bottlenecks

- **Harsh Environment:** The computer must survive constant, rigorous shock and vibration from the road, which can loosen components and cause system failure.
- **Unstable Power:** Vehicle electrical systems are notoriously “dirty,” with power fluctuations during startup and shutdown that can corrupt data.
- **Computational Demand:** Our system requires significant computing power to handle the data stream from the MotionCam-3D, often for real-time operations like 3D Instant Meshing.
- **Connectivity:** The solution must provide stable power and data connections for our camera, inputs for other vehicle sensors, and robust wireless communication (5G/4G) for real-time data uploading.

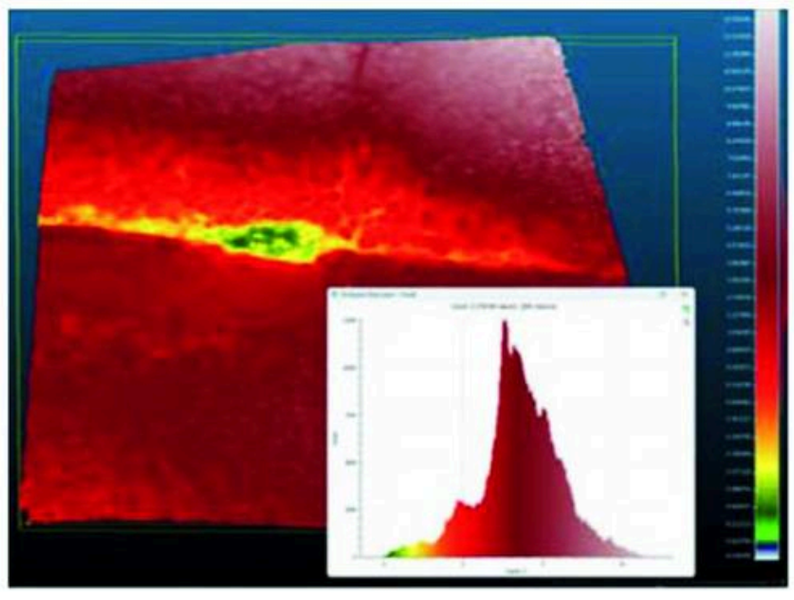
Freezing Motion with Parallel Structured Light For Road Inspection



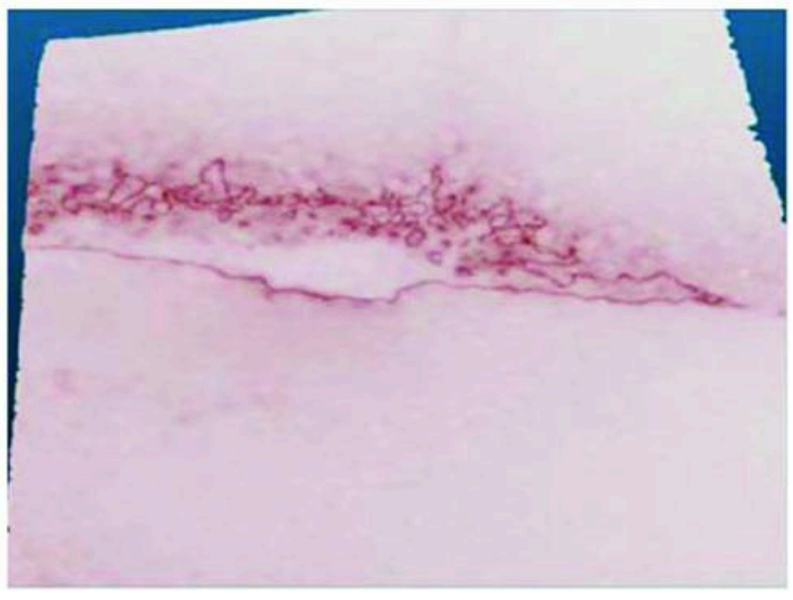
3D View



Depth Hue Z



Depth View



Normal View

Traditional scanning causes motion blur on moving vehicles because it requires multiple frames to build a model.

The MotionCam-3D Color M overcomes this using proprietary Parallel Structured Light (PSL) technology.

By capturing all depth information in a single exposure, the system "freezes" motion to deliver high-resolution 3D reconstruction of pavement even at normal driving speeds.

Our Technology

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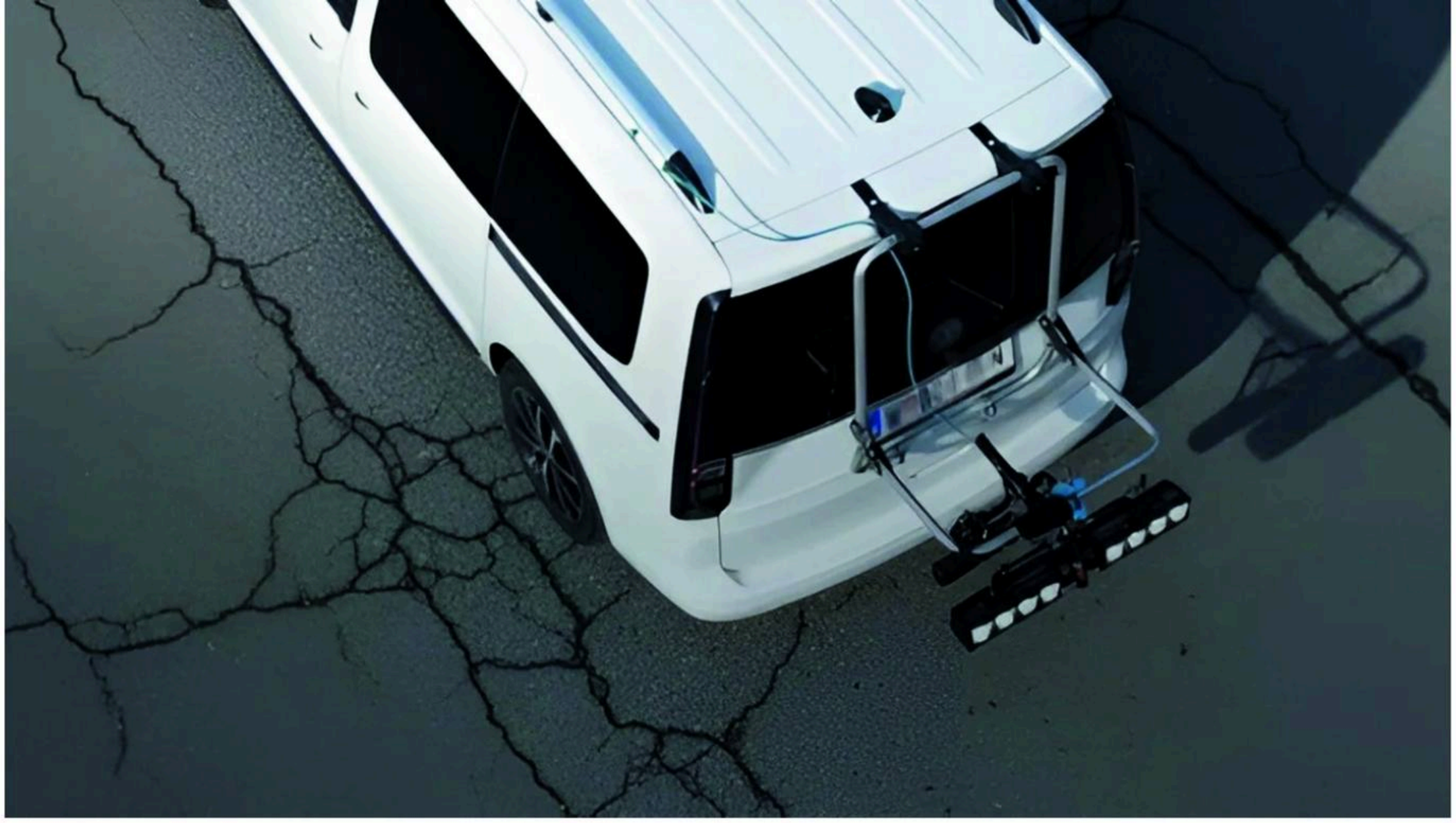


The Solution

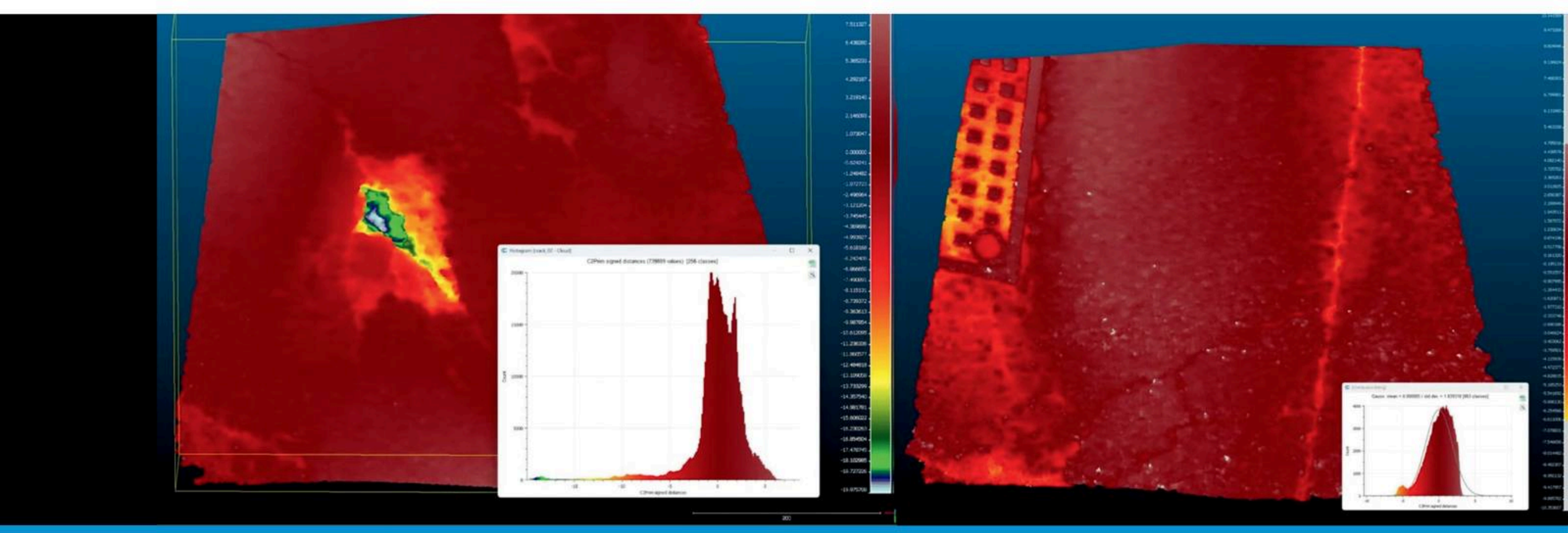
Tech Selection

To create a field-ready solution that their customers can depend on, Neousys needed a 3D vision partner whose technology could deliver industry-grade scanning precision in the chaotic environment of a moving vehicle.

They selected Photoneo and our MotionCam-3D technology to serve as the high-performance sensor at the heart of their rugged computing platform.

The platform directly addresses every challenge we faced in designing the mobile inspection solution:

- **Vibration Immunity:** With E-MARK certification and compatibility with MIL-STD-810G, the computers can easily withstand the shock and vibration (up to 3Grms) of a moving vehicle.
- **Reliable Performance:** These systems support powerful Intel® Core™ processors and dedicated NVIDIA® graphics cards. This ensures our 3D meshing libraries can run smoothly on-the-fly.
- **Simplified Connectivity:** The Nuvo's integrated PoE+ ports are a perfect match for our MotionCam-3D Color M, allowing us to power the camera and transmit data through a single, secure cable.
- **Intelligent Power Management:** The most critical feature for data integrity is Neousys' patented SuperCAP power backup module. This module monitors the vehicle's power. If it detects a failure or shutdown, it instantly provides backup power to the computer.
- **Real-Time Data Upload:** With mini-PCIe sockets and SIM slots, the system is easily equipped with 5G/4G or WiFi modules.



Safer Roads,
Lower Costs

80% saved on manual inspection time

95% accurate prediction of road condition

50% saved on maintenance costs

Resistance to harsh environments

Wide range of defects covered

Key Benefits

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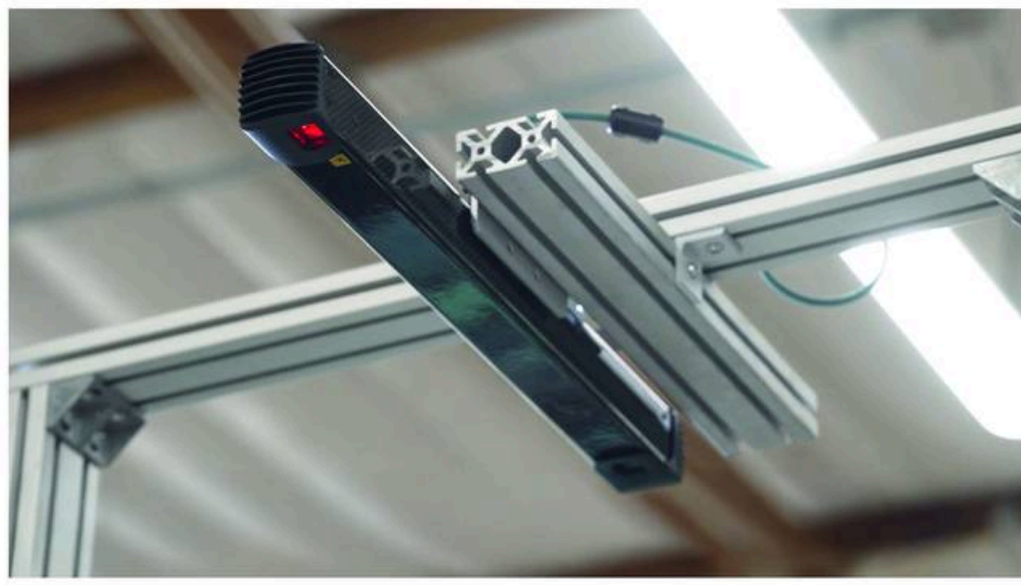
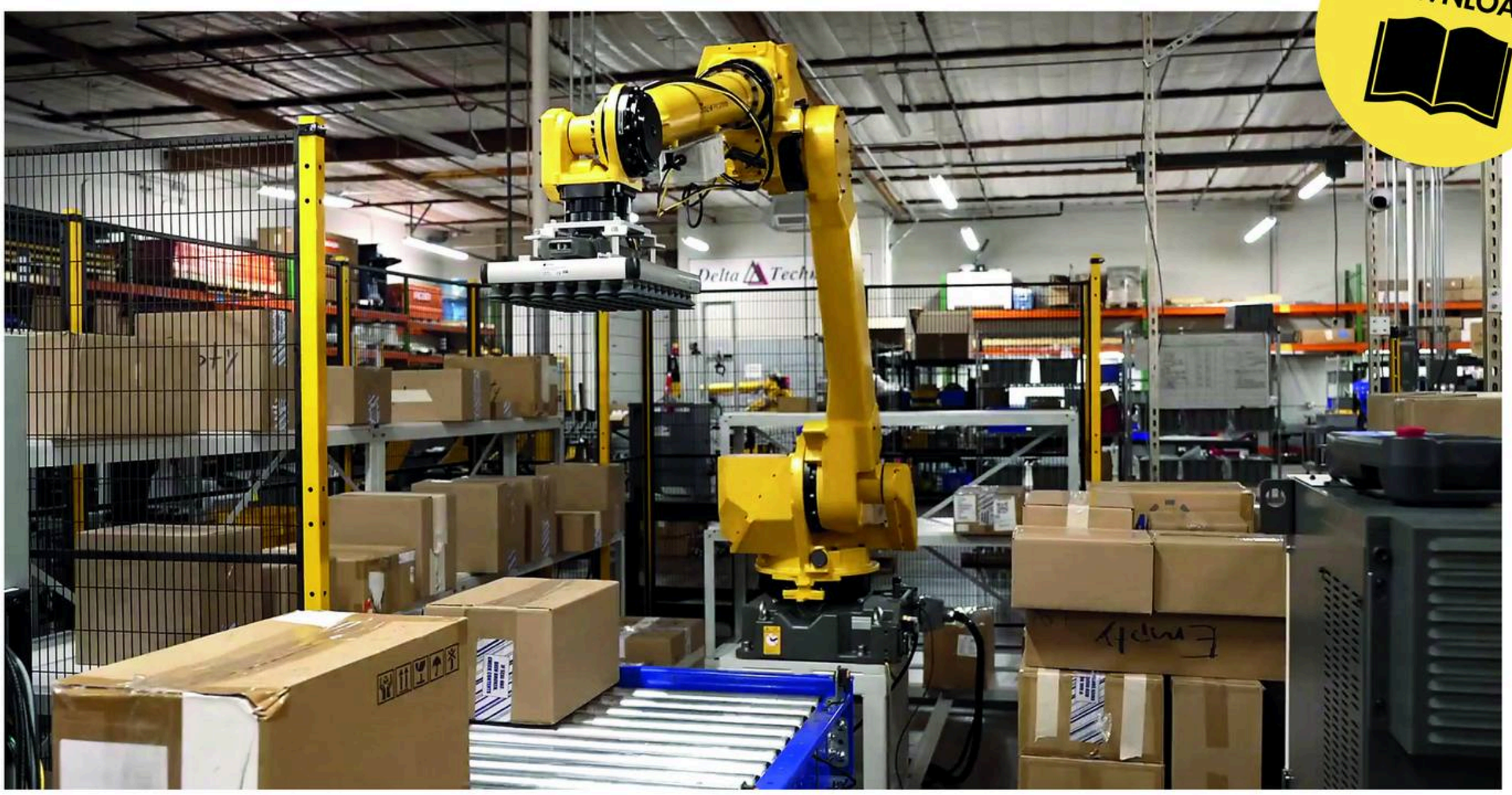
Watch the video

Enabling Real-Time Mixed-Case Palletizing



Sim-to-Real Accuracy With 0% Error

FREE DOWNLOAD



USE CASE

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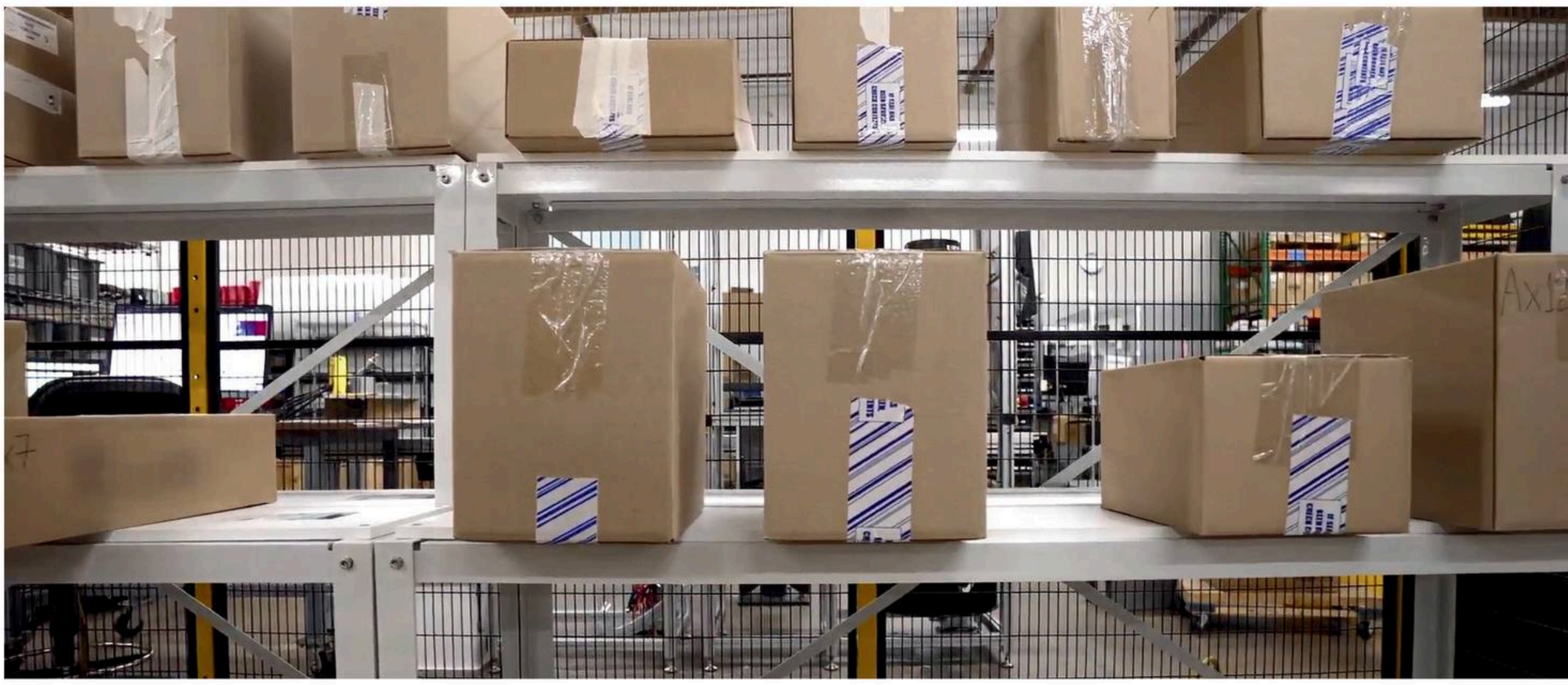
Overview

- Challenge: Automating a chaotic, injury-prone manual palletizing line handling random, unsequenced parcels.
- The Missing Link: High-fidelity, real-time 3D perception to "see" incoming freight and verify outgoing stability.
- Vision Technology: Photoneo MotionCam-3D.
- The Partnership: Photoneo (Vision) + Jacobi Robotics (Software/Brain) + Delta Technology (Integration) + FANUC (Motion).
- Result: A live production cell with 100% pallet stability and detection of damaged goods before they enter the stack.

Client: Defense Manufacturer via Jacobi Robotics & Delta Technology
Industry: Logistics
Application: Mixed-Case Palletizing
Technology: MotionCam-3D



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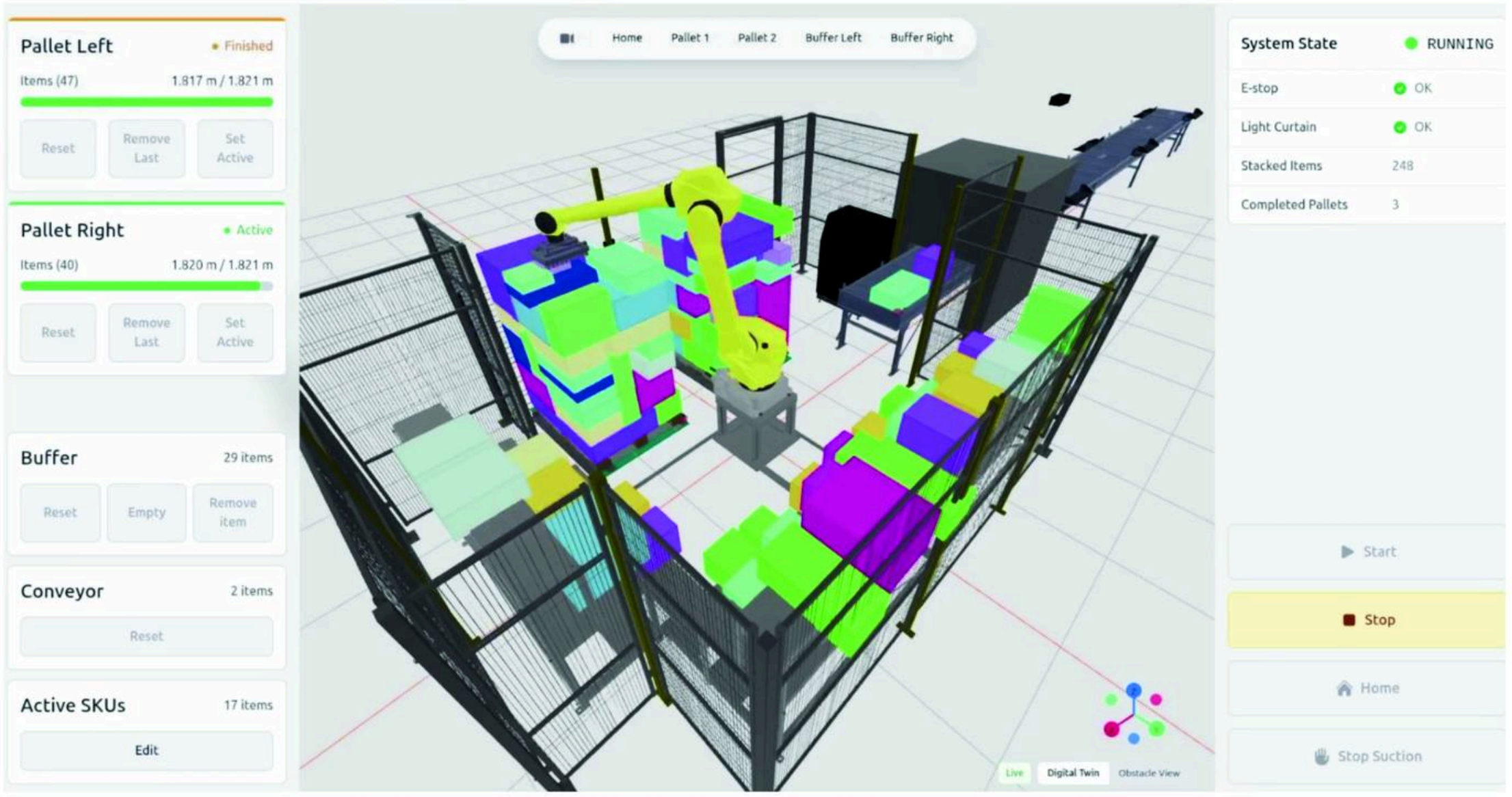

The Challenge

You Can’t Stack What You Can’t See

A major defense manufacturer faced a risky bottleneck at its shipping dock: operators manually palletizing random, unsorted cases of varying sizes and weights (up to 27.2 kg/60 lbs).

While Jacobi Robotics and Delta Technology provided the integration and the planning "brain," standard blind robots couldn't handle this unpredictable chaos.

Because the incoming sequence changed every second without a fixed recipe, the system required high-speed "eyes" capable of capturing the conveyor's reality in milliseconds.

MotionCam-3D as the Source of Truth

To solve this, the cell was equipped with Photoneo's MotionCam-3D.

Unlike traditional static scanners that require the conveyor to stop for a snapshot, MotionCam-3D captures high-resolution 3D point clouds of objects in motion. This allowed the system to feed precise, real-time data into Jacobi's OmniPalletizer software without slowing down throughput.

The vision system performs two distinct, mission-critical roles in this workflow:

The Solution

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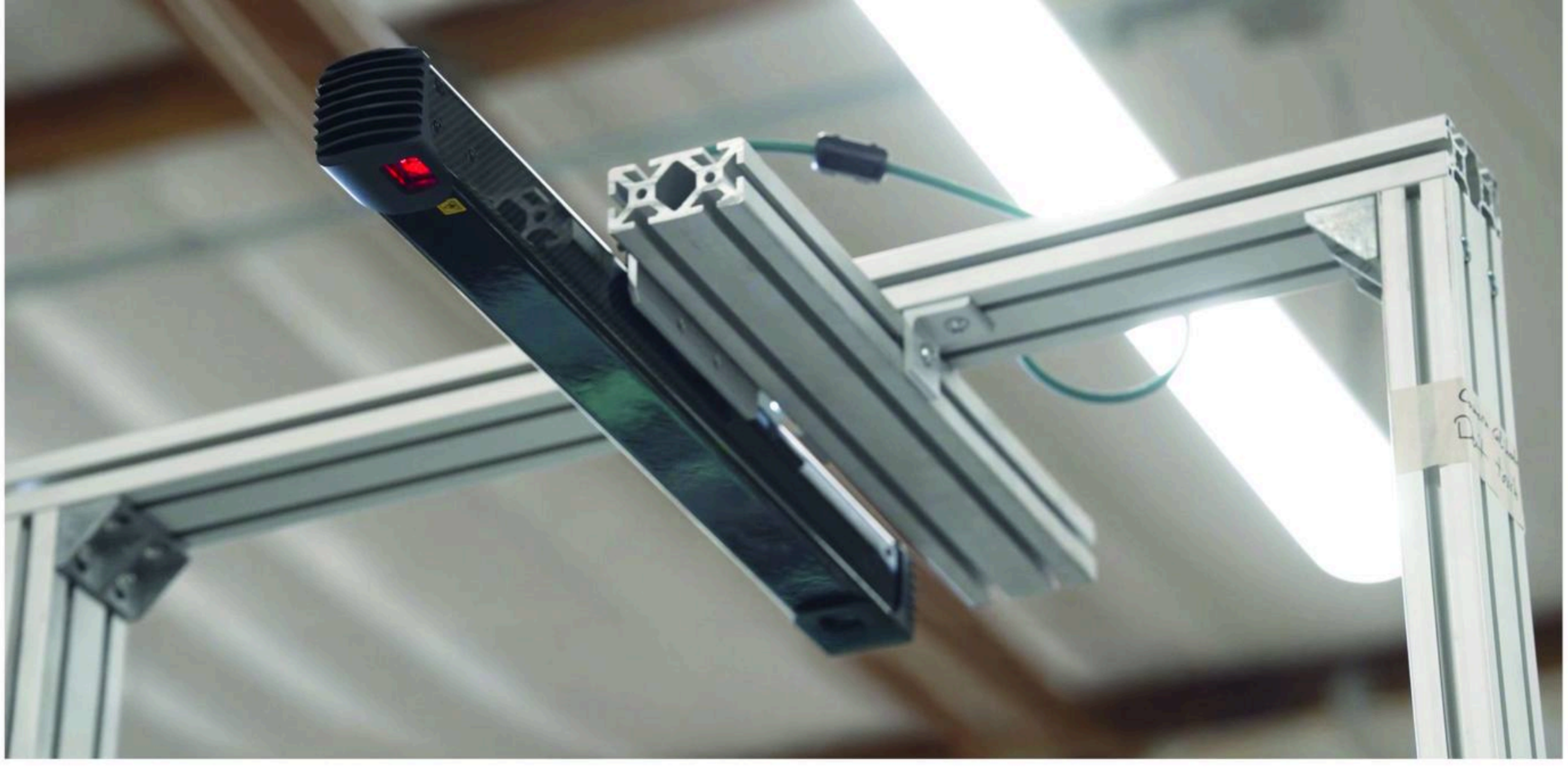
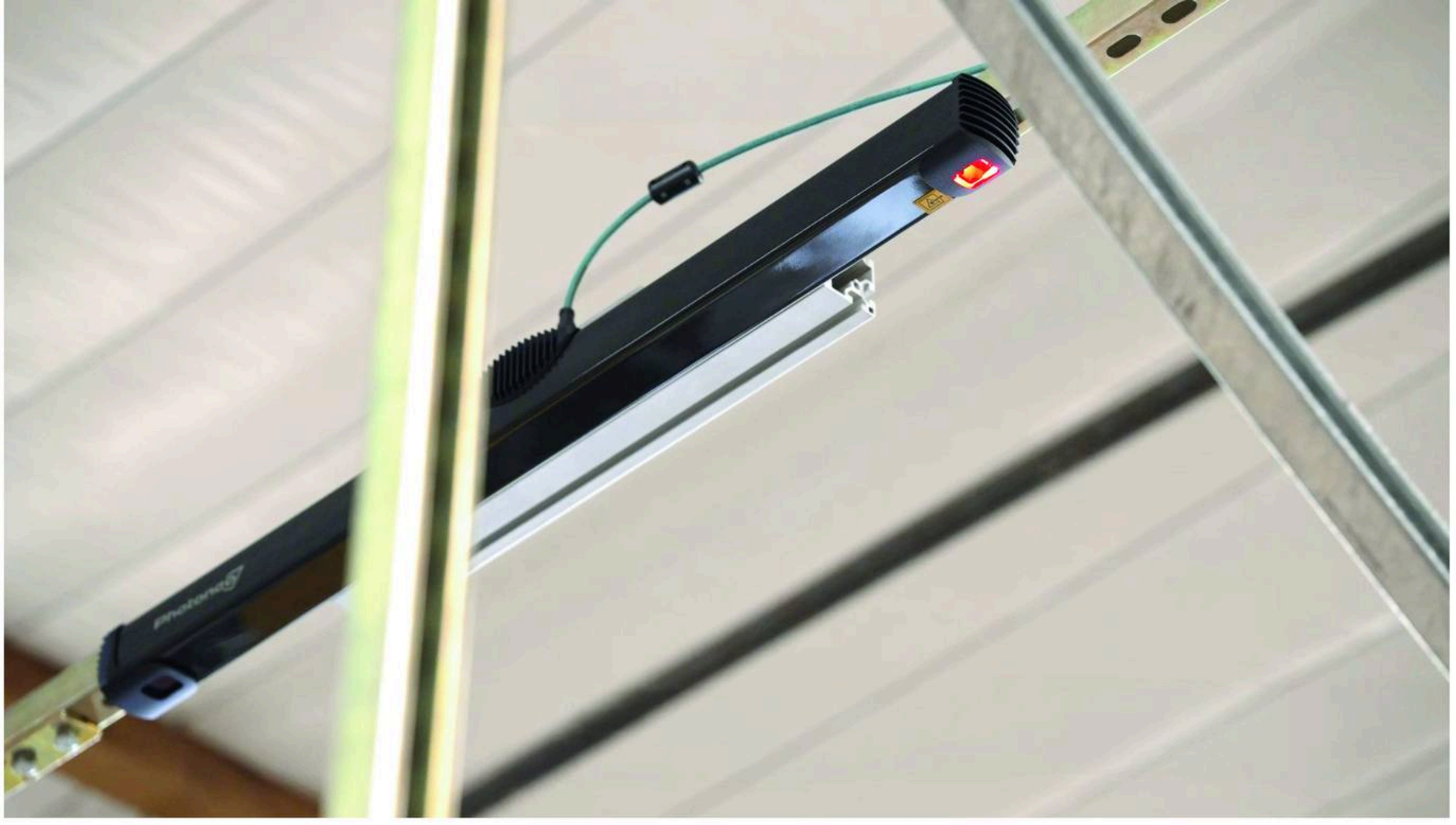


The Infeed Scanner

Perfect Perception in Motion

As cases flow down the conveyor, the MotionCam-3D acts as the gatekeeper.

- **Real-Time Dimensioning:** The scanner instantly captures the exact dimensions (L x W x H) and orientation of every box. This data is fed to Jacobi's path planner, allowing the software to solve the "Tetris" puzzle of where the box fits best on the pallet.
- **Quality Control:** The high-fidelity 3D data detects crushed corners, open flaps, or structural damage before the robot picks the box. Damaged boxes are flagged and rejected, preventing unstable stacks and ensuring load integrity for freight carriers.



The Verification Scanner

Closing the Loop

Placement accuracy is everything in mixed-case palletizing. Once the robot places a case, a secondary visual check confirms the reality matches the plan.

- **Placement Verification:** The system verifies that the box was placed in the correct coordinates and orientation.
- **Stack Stability Check:** By scanning the building layers, the system confirms the pallet remains flat and stable, ensuring 100% stability even as the stack grows higher.

Why the Partnership Works

Even the most advanced AI planner cannot build a stable pallet if its input data is wrong. Photoneo ensures that Jacobi's 'brain' receives a perfect picture of reality.

- **Photoneo:** Provides the "Eyes" (High-quality 3D data & QC).
- **Jacobi Robotics:** Provides the "Brain" (Real-time path planning & physics-aware logic).
- **FANUC:** Provides the "Muscle" (Reliable industrial robotics).
- **Delta Technology:** Delivers the "Body" (Seamless integration into the warehouse floor).

The Partnership

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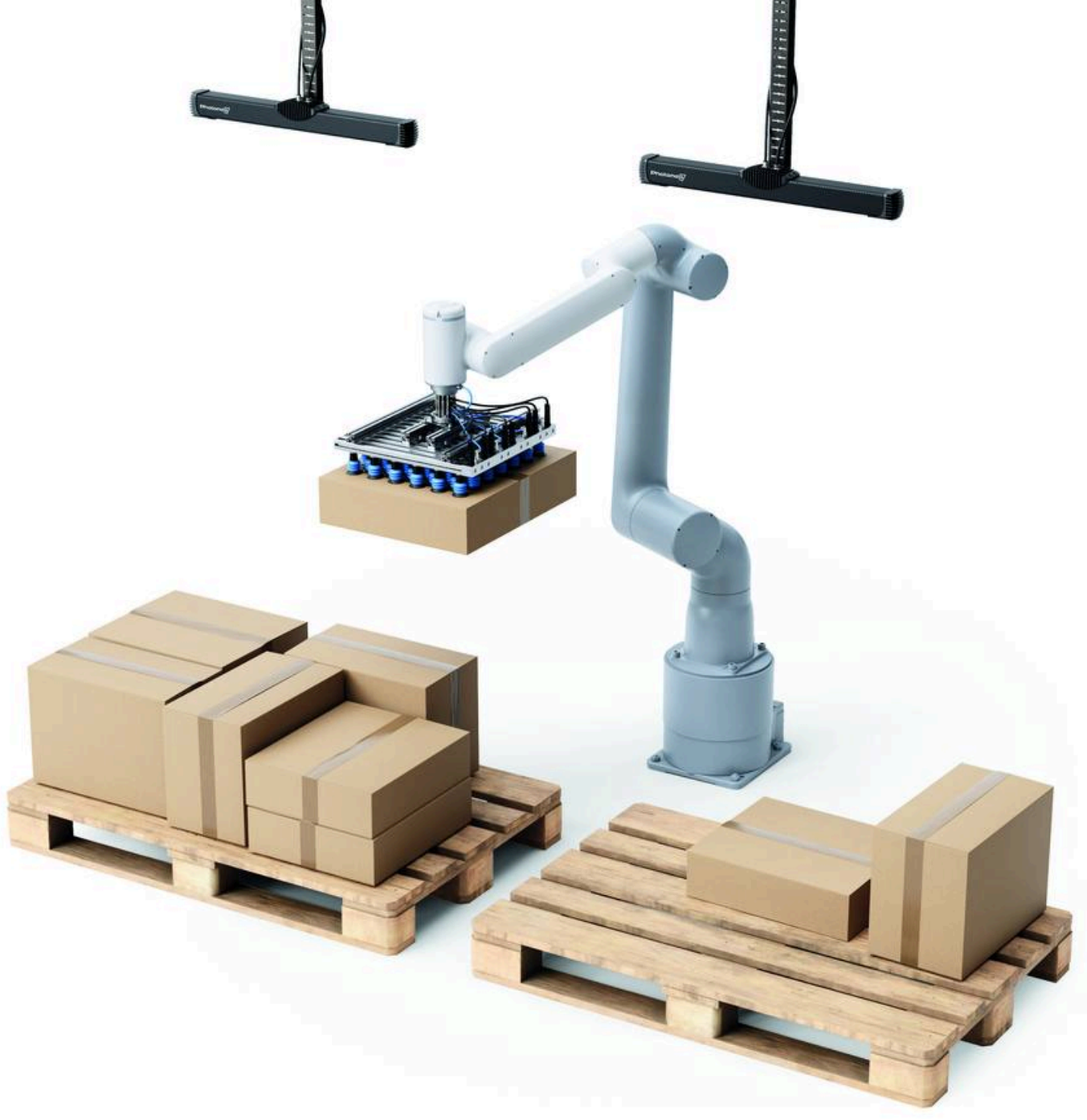
Validated, Safe, and Scalable



100% Stability: Vision-verified stacking ensures no product loss.

Zero Downtime for "Surprises": Damaged boxes are seen and stopped.

Precise dimensioning allows Jacobi to pack pallets up to 90% density

3D data accuracy allowed for simulation to match real-world cycle times with 0% error.

The Results

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